Tlhware

Buka e e tsamaelanang le OctoStudio



Diporojeke tse 20. Ditshwantsho tsa gago. Lentswe la gago. Khoutu ya gago.

Circuit Kids

Rea go amogela mo Tlhwareng

Sediriswa le sedirisiwa sengwe le sengwe mo Circuit Kids se tshwanetse go nna le buka e e se tsamaelanang. Buka eno e dira gore molao oo e nne tsela ya bafelegi: pele o itse tsela mo app, morago o dire sengwe se se nang le bokao.

Ka ga kgatiso eno

- Babadi ba ba reretsweng: dingwaga di le 7-12, ka molekane kgotsa motho yo o godileng fa go tlhokega. Bana ba bannye ba ka nna le seabe ka go taka, go anela le go tshameka ka thuso ya motho yo o godileng.
- Kgang: ditsala tse di akanyeditsweng Neo, Kagiso le Lesedi, le nkuku wa bone, ba ya mo kgaolong ya Bokone Bophirima. Tlhware ya pampiri e tsamaya le bone go tloga kgang ya ntlha go ya kwa ya bofelo. Kgang ya 1 e simolola kwa Leseding la Molemane. Mafelo a nnete le ngwao tse di tshelang ga di tshwantshiwe e le dilo tsa batho ba ba akanyeditsweng.
- Thulaganyo: karolo ya go tsamaya mo app, morago diporojeke tse 20: kgang le go dira, morago khoutu le go leka. Porojeke 20 e na le karolo e nngwe ya go leka le go akanya.
- Sebopego sa buka: e dirilwe go balwa mo mogaleng kgotsa mo tablet, kgotsa go gatisiwa go ithuta mmogo.
- Leina le laesense: Tlhware ke lefoko la Setswana la noga ya python. Buka eno e na le laesense ya CC BY-NC-SA 4.0; bona tsebiso ya ditshwanelo tsa kopi le laesense ka fa tlase.

Buka eno ke eng, mme ga se eng

Thulaganyo ya ntlha ya go ruta e ne ya tlhatlhobiwa ka kaedi ya semmuso ya July 2026 ya OctoStudio 1.1.0 le ya godimo. Diporojeke tseno tsa ntlha ga di ise di dirwe mo disedirisweng tsa nnete kgotsa le bana. Dipholo tse di solofelwang ke dikgopolo tsa go leka, e seng dikopanelo tsa gore teko e fetile.

Circuit Kids e mo go kwadisweng. Dipotso: 072 123 1490 | info@circuitkids.co.za.

Ditshwanelo tsa kopi le laesense

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O ka abelana le go fetola dilo tseno ka bolokologi fa e se go dira madi. Bolela Circuit Kids, golaganya laesense mme o tshwaye diphetogo dipe. Abelana dipalo tsa gago ka laesense e e tshwanang.

Diteng

Bala kgang pele. E tlogele e nne le wena. Morago dira selo se le sengwe sa lefatshe leo se tsamaye, se bue kgotsa se araba.

Bona dinao tsa gago

Pele ga Porojeke 1: baakanya app, itse dikirini, dira setaki sa gago sa ntlha, ithute go baakanya phoso, mme o boloke kopi e e sireletsegileng ya tiro ya gago.

Diporojeke tse 20

01 Dumela kwa Leseding - Go tobetsa, sprite, polelo

02 Pele tsela, morago metsi - Motsamao le tatelano

03 Mokgwa mo lerakeng la ga Nkuku - Go taka le go boeletsa

04 Morula e tsaya nako ya yone - Bogolo le go ema

05 Lefika le le ntseng le panya - Fitlha, bontsha le disprite tse pedi

06 Nkuku o dumela moranodi - Maekerofouno le modumo

07 Mang o araba eng - Ditaki tse di farologaneng tsa ketsakgato

08 Karata ya poso go tswa Taung - Khamera le go sega ditshwantsho

09 Kgaswane e e tsamayang - Diaparo le motsamao o o bonalang

10 Dikganya tse tharo fa godimo ga lekgwa - Metheo le ditiragalo

11 Botlhe ba a nna, botlhe ba a bua - Melaetsa le go tsaya dikhutlo

12 Thaba e fetoga ditlhase tsa mollo - Kganya le mmala

13 Seatla se se sa kgoneng go šikinyega - Šikinyega, le tsela e nngwe ya go tobetsa

14 Mokoloko wa dithaba tsa bogologolo mo fatshe ga kicheng - Go sega le ketsakgato ya go ama

15 Dikgakana tse tharo kwa morago ga tanka - Dibolokwa tse di fetogang le dintlha

16 A botlhe ba tharo ba gae? - Ditlhokego le sepheo

17 Ledimo le le tlileng go tswa kae - Dipalo tse di sa rulagangwang

18 Lefoko la mokoloko ka bone - Dibolokwa tse di rulaganyeditsweng

19 Kgang ya motho yo o kgakala - Vidio, GIF le faele ya porojeke

20 Se re se isang kwa gae - Itlhame kgang ya gago e e dirisanang

Motho yo o godileng o re thusa go simolola

Pele ga bana ba tla

- Kenya OctoStudio go tswa mo dikgokaganyong tsa semmuso tsa lebenkele la di-app kwa octostudio.org ka inthanete. Ntšhafatsa pele ga sekolo; diaparo di tlhoka kgatiso 1.1.0 kgotsa e ntšhwa. Bula app lekgetlo le le lengwe mme o lekole puo e e tlhophilweng.
- Dirisa mogala kgotsa tablet e e tshegeditsweng: kaedi ya gompieno e kwala Android 8+, iOS 15+, kgotsa Chromebooks tse di nang le di-app tsa Android. Dikhomputara tse dingwe tsa tlwaelo ga di tshegediwe ke app eno. Lekola disedirisiwa tsa sekolo tsa nnete pele o solofetsa kharikhulamo.
- Morago ga go kenya, go tlhama go dira ntle le inthanete. Boloka sebaka sa ditshwantsho, modumo le difaele tsa porojeke. Go romela faele ka inthanete go ka tlhoka data; go tlhama ntle le inthanete ga se go abelana mahala ka inthanete.
- Dira babadi ka bobedi: mongwe ke moetapele yo o tobetsang sekirini, mongwe ke moetapele yo o balang kgotsa a supang. Fetolang porojeke nngwe le nngwe. Ka fa tlase ga dingwaga tse 7, dirisang go taka, go anela le motshameko wa boloko jo bo le nngwe o o kaelwang ke motho yo o godileng go na le go solofela tatelano yotlhe ka bone.
- Mo Gae > Disetingi, bula Matshwao a Tshimologo a bafelegi. Bontsha go tobetsa sekirini go thusa ditshupo. Leka VoiceOver kgotsa TalkBack ya sediriswa le medumo ya dibolokwa tsa khoutu le moithuti yo o di tlhokang. Leka gore o bone gore go kgonega, o se ka wa akanya gore setting e le nngwe e a lekana botlhe.

Kano ya rona ya polokesego

Dirisa ditshwantsho le medumo ya lebokoso la dibuka pele. Kopa pele ga o tsaya setshwantsho sa motho kgotsa o rekota lentswe la gagwe. O se ka wa akaretsa maina a dikolo, diaterese, dinomore tsa mogala kgotsa dipuisano tsa poraefete mo diporojekeng. Motho yo o godileng o sekaseka sengwe le sengwe se se tlogang mo sedirisweng. O se ka wa dirisa app eno fa o tshela tsela. Tshwara disediriswa ka go tiya fa godimo ga selo se se bonolo fa o sega kgotsa o šikinya. Ga go na go šikinya ka maatla.

Pholisi ya poraefete ya app e re data e bolokwa mo selegaeng mme ga e kgobokangwe ke OctoStudio. Faele e e romelwang kwa tirelong e nngwe ke tshenolo e e ikemetseng. Boloka kopi pele ga go tlosa app; go tlosa go ka phimola data ya selegae ya app. Ga go tlhokege go abelana mo phaposing gotlhe go fetsa buka eno.

Tsaya OctoStudio mahala

Skena khoutu ya sediriswa sa gago ka khamera, kgotsa o kwale aterese ya webosaete e e kwa tlase. App e dirilwe ke setlhopha sa MIT Lifelong Kindergarten (setlhopha sa Scratch) mme dikgokagano di tswa kwa octostudio.org.

Android (Google Play):
https://play.google.com/store/apps/details?id=edu.mit.octostudio

iPhone le iPad (App Store):
https://apps.apple.com/app/octostudio/id6461573316

Itse dikirini tse pedi

Gae: shelofo ya diporojeke tsa gago

Gae e kwala diporojeke tsa gago. Dira porojeke e ntšhwa e simolola e ntšhwa. Sekaseka e na le dikao tse o ka tshamekang ka tsone. Lekgasa le bula Disetingi. Lenaneo la porojeke le go naya Kopa, Rata, Abelana le Phimola. Leka Kopa mo porojekeng ya go ikatisa, e seng Phimola mo tirong ya gago e e botoka.

Moeditara: lebala la gago le lennye

SETHALE: bona le go ama kgang ya gago

Play / Stop Okeletsa ka bonako: sprite, motheo, modumo

LENANEO LA SPRITE: tlhopha khoutu ya mang e o e fetolang

SEFELO SA KHOUTU: kgomaretsa ditaelo tsa gago

LENANEO LA DIBOLOKO: batla ditaelo

Mmapa o o nolofaditsweng wa ntlha, e seng setshwantsho sa sekirini. Mo Chromebooks le di-tablet tse dikgolo dikarolo tse di tshwanang di ka rulaganngwa fa thoko ga nngwe.

Sprite ke sebopego kgotsa selo se o ka se kwalelang khoutu. Motheo ke lefelo le le kwa morago ga yone. Sethale ke felo fa porojeke e dirang gone. Boloko ke taelo e le nngwe. Setaki, kgotsa skripte, ke setlhopha se se golaganeng sa dibolokwa tsa sprite e le nngwe.

Tobetsa sprite mo lenaneong la disprite pele ga o e kwalela khoutu. Sethale se bontsha kgang yotlhe, fela sefelo sa khoutu ke sa sprite e e tlhophilweng. Fa sebopego se se fosagetseng se tsamaya, lekola kgetho pele.

Dipuo tsa Aforika Borwa mo OctoStudio

Lenaneo la gompieno la dipuo tsa OctoStudio le akaretsa Afrikaans, English, isiXhosa, isiZulu, Sepedi (Northern Sotho), Setswana (Tswana) le Tshivenḓa (Venda). Tlhopha puo ya app mo Gae > Disetingi. Baithuti ba ka anela dikgang ka dipuo tse ba di dirisang mo gae; kopa motho yo o di buang gore a tlhatlhobe dikgumedišo le direkoto.

Mothopo: https://octostudio.org/about | Lenaneo la dipuo le lekolotswe ka 10 Diphalane 2026.

Dira taelo ya gago ya ntlha

- Dira porojeke e ntšhwa go tswa mo Gae. Tlhopha motheo go tswa mo lebokosong la dibuka. Dirisa Okeletsa sprite mme o tlhophe emoji kgotsa Penta sebopego. Boloka sprite e le nngwe fela mo go ikatiseng.
- Tlhopha sprite ya gago mo lenaneong la disprite. Batla mofuta wa "Go simolola leng?" mo lenaneong. Goga boloko jwa Play-simolola mo sefelong sa khoutu.
- Bula Motsamao. Goga boloko jwa Tlola ka fa tlase ga boloko jwa go simolola go fitlha dikarolo di kgomarela. Tobetsa palo ya yone mme o leke 20.
- Tobetsa Play ka fa tlase ga sethale. Sprite ya gago e tshwanetse go tlola. Tobetsa Stop. Jaanong fetola 20 go nna 40 mme o tobetse Play gape. Go fetogile eng?
- Goga boloko jwa bobedi jwa Tlola ka fa tlase ga jwa ntlha. Akanya gore dithoko tse pedi di tla lebega jang. Tshameka mme o bapise. Fa go sa direge sepe, golaganya dibolokwa gape mme o lekole sprite e e tlhophilweng.

BALA O ya TLASE MO SETAKING

Fa Play e simolola

Tlola 20

Tlola 40

Khoutu ke leano le sediriswa se ka le latelang. Boloko jwa go simolola bo araba "Leng?" Dibolokwa tse di latelang di araba "Dira eng?" Go fetola palo e le nngwe go fetola leano. Go leka ke go leka leano le go lebelela ka kelotlhoko.

Diresipi mo bukeng eno di dirisa mafoko a tlhaloso a dibolokwa le dikgetho tsa tsone tsa lenaneo. Ga se khoutu e e kwadilweng ka tlhamalalo, ditshwantsho tsa app kgotsa kano ya gore puo nngwe le nngwe e dirisa maina a a tshwanang. Dirisa kaedi ya semmuso go tshwantsha ditshwantsho tsa dibolokwa. Mela e e tsentsweng e tsena mo molomong wa loop kgotsa tlhopho.

Lebokoso la gago la didiriswa tsa dibolokwa

Go simolola leng?

Play, go tobetsa, go šikinya, go ama sprite, go fetola motheo. Ketsakgato e bolelela setaki gore se simolole leng.

Motsamao

Tsamaya, tlola, retolola, ya kwa felong la sethale, lebelela kwa ntlheng. Dirisa go tsamaya go gonnye pele ga go golo.

Mafoko le Medumo

Bubble ya puo, mokwalo wa sethale, lentswe le le buang, modumo o o rekotilweng kgotsa wa lebokoso la dibuka. Letsa modumo "go fitlha o fela" fa tiro e e latelang e tshwanetse go ema.

Lefelo

Bogolo, diaparo, fitlha/bontsha, motheo, kwa pele/kwa morago. Diaparo ke ditebego tse di farologaneng tsa sprite e le nngwe.

Mebala le Lesedi

Phatsima, diphetogo tsa mmala, lebone. Buka eno e dirisa lesedi la sekirini gore lebone le se tlhokege.

Taolo

Boeletsa, ka metlha, fa, ema, emisa, romela le go amogela molaetsa. Dibolokwa tseno di rulaganya tatelano.

Dibolokwa tse dingwe

Sega, dibolokwa tse di fetogang, dipalo, beam le dibolokwa tse di ikgethileng. Okeletsa bonale fela fa kgang ya gago e di tlhoka.

Dibolokwa dingwe di na le dikgetho tsa lebelo: bonya, magareng, bonako le ka bonako-bonako. "Baya bogolo" e tlhopha bogolo jo bo feletseng; "fetola bogolo" e okeletsa kgotsa e tlosa go tswa mo bogolong jwa jaanong. "Baya dintlha" e emisa palo; "fetola dintlha" e okeletsa mo go yone. Dipharologano tseno di a tlhokega fa o simolola gape.

Boloka tiro ya gago. Baakanya diphoso tsa gago.

Porojeke ke tse dintsi go feta vidio

- Fetsa phetogo e nnye, tobetsa Stop, morago o dirise Gae. Batla porojeke ya gago mo lenaneong la Gae mme o e bule gape. Lekola gore tiro ya gago ya bofelo e santse e le mo sedirisweng seno.
- Pele ga teko e kgolo, dirisa lenaneo la porojeke > Kopa. Bula kopi mme o e fetole. Boloka ya ntlha e bolokegile.
- Go dira kopi ya go boloka, bula Abelana > Romela Faele ya Porojeke. Tlhopha kgetho ya go boloka faele mo sedirisweng fa e le teng. Boloka kopi mo foldareng e motho yo o godileng a ka e bonang. Kaedi e tlhomamisa go boloka faele ya porojeke mo sedirisweng; dikonope tsa letlapa la go abelana di a farologana.
- Leka kopi: bula faele eo ya .octostudio e e bolokilweng ka OctoStudio mme o lekole disprite le dibolokwa tsa yone. Go amogela faele ya porojeke go tlhoka OctoStudio. Vidio kgotsa GIF ke rekoto fela, e seng khoutu e e ka baakanngwang.

Fa sengwe se tsamaya ka phoso

Ga go dire sepe? Lekola boloko jwa go simolola, kgokagano e e kgomarelang le ketsakgato e e siameng. Sprite e e fosagetseng? E tlhophe mo lenaneong la disprite. Dibolokwa di nyeletse kwa ntle ga sekirini? Kgutshwafatsa kgotsa o rulaganye sefelo sa khoutu. Phetogo e e fosagetseng? Leka Busetsa morago, morago Dira gape fa go tlhokega. Sprite e e sa bonaleng? Lekola Fitlha le bogolo. Modumo o o didimetseng? Lekola kgetho ya modumo, bophara le ditetla. Fa ketsakgato e fetola morago ga go tshameka nngwe le nngwe, busetsa boleng jwa yone jwa tshimologo mo setakeng sa Play.

Ema pele ga go busetsa sprite kwa tshimologong. Tobetsa Play gape go leka sešwa. O se ka wa phimola porojeke go baakanya phoso e nnye. Bolelela molekane wa gago: "Ke ne ke solofetse ___. Ke bone ___. Ke tla fetola ___."

Ikatise go boloka kopi ntle le ditshwantsho tsa poraefete kgotsa medumo. Abelana mo phaposing fela fa motho yo o godileng a dumela. Go boloka faele e sireletsegile ga go tlhoke go e baya mo inthaneteng.

PROJEKE 01

Dumela kwa Leseding



Neo o ne a builwe gore metsi a kwa Leseding la Molemane ga a ke a ema. A tswa mo ntle ga lefatshe, a phepa ka fa o ka balang maje. O ne a eme kwa mothorong wa metsi, a tshwere bukana ya gagwe ya go taka thata. Kagiso le Lesedi ba ne ba setse ba tshela diatla mo metsing. Ba ne ba itse lefelo leno botshelo jotlhe jwa bone. Neo o ne a le itse fela a le mo setulong sa morago sa koloi.

Nkuku a re, "Dumela, Neo," a bua le metsi jaaka e kete ke motho. Bana ba tshega, fela ka kelotlhoko. Neo a bula bukana ya gagwe. O ne a taka tlhware, e telele, e tala, e kopame jaaka potso. Nkuku a dumalana a re, "Tlhware. Ke lone lefoko la rona la noga eo." Neo a le boeletsa. Le ne le ikutlwala jaaka go apara seeto se seša.

Morago, kwa tafoleng, Lesedi a thaba-thaba setshwantsho. "A e bue fa o e tobetsa." Neo a akanya. Ga a ka a nna le puo e telele ya tlhware. O e file lefoko le le lengwe fela le a neng a le tshaba go le bua kwa kgorong. "Dumela."

Potso ya kgang: *Tlhware ya gago e ka reng gore motho a ikutlwe a amogelesegile?*

Bokgoni: Go tobetsa, sprite, polelo

Dira

1. Dira porojeke e ntšha mme o tlhophe motheo o o idibetseng wa lebokoso la dibuka. Taka sprite ya tlhware e e rategang ya pampiri. Mmele o o kobegileng o bonolo le sefatlhego di lekane; ga go tlhokege setshwantsho sa phologolo ya naga.
2. Tlhopha sprite ya gago mo lenaneong la disprite. Goga boloko jwa Go tobetsa-simolola mo lefelong la khoutu. Tlhopha kgetho e e

kayang go tobetsa sprite eno, e seng felo fa e leng teng.

3. Mo go Mafoko le Medumo, tsenya boloko jwa bubble ya puo. Kwala 'Dumela, o amogelesegile' mme o beye nako ya metsotswana e le 2.

Khoutu (recipe)

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TLHWARE YA GAGO YA PAMPIRI YA GO DUMEDISA
When this sprite is tapped
Say in bubble: Dumela, welcome for 2 seconds
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Go dira jang

Go tobetsa ke ketsakgato. E simolola setaki seno fela fa sprite e e tlhophilweng e tobeditswe. Bubble ke tiro. Ga go na sepe se se tshwanetseng go dirwa pele ga go tobetsa.

Leka

Tobetsa Play fa go tlhokega gore o tsene mo pono e e dirang. Tobetsa sprite, ba ba tobetse karolo e e senang sepe ya motheo. Ke sprite fela e e tshwanetseng go dumedisa. Lekola kgetho ya go tobetsa fa tsotlhe di arabela.

Fetola

Kwala dikgumedišo mo pampiring ka dipuo tse lelapa la gago le di dirisang. Tlhopha nngwe ya sprite ya gago. Kopa motho yo o buang puo eo gore a tlhatlhobe mafoko; o seka wa ikanya modumo wa tsamaiso fela.

Tlhalosa le go abelana

Bontsha molekane wa gago gore o tlhopha sprite jang le go bona khoutu ya yone. Boloka kopi ya faele ya porojeke morago ga go dumedisa ga ntlha go atlegang.

POROJEKE 02

Pele tsela, morago metsi



Kombi e tswang Mahikeng e ne e tlala ka bonya. Kagiso o ne a bala ditlhogo. Lesedi a bala dikgetsana. Neo a bala metsotso, ka gonne o ne a batla go setse a le kwa metsing. Nkuku fela o ne a ikhutsa mo setulong sa gagwe. "Dilo di na le tatelano," a bua. "Pele tsela, morago felo."

Tsela e ne e tsamaya e sididi fa pele ga dikgoro le ditlhare tsa mooka. Morago ga dikilometara di se kae, kaloi ya tonki kgotsa mosimane yo o tsamaisang baesekele o ne a ema mo thoko ga yone. Tlhware ya pampiri ya ga Neo e ne e tshologela mo lengwapeng la gagwe ka fa bokgoreng bongwe le bongwe. O ne a leka go taka tsela mme a retelelwa. A simolola gape ka tshimologo le bokhutlo. Ntlo e nnye kwa letlhakoreng le lengwe. Metsi a mannye kwa go le lengwe. Gareng ga one, tsela e khutshwane e a e kgonang go e tsamaya ka dikgato.

Fa a tsamaisa tlhware pele a taka tsela, e ne ya nanabela mo pampiring e e se na sepe. Fa a taka tsela pele, e ne ya nna le kwa e yang teng. Kagiso a lebelela fa lehetleng la gagwe. "Tatelano e a tlhokega," a bua, a utlwala jaaka Nkuku. Neo a tshega mme a se ka a gana.

Potso ya kgang: *Tlhware ya gago e tshwanetse go simolola le go bofelela kae mo loetong lwa yone?*

Bokgoni: Motsamao le tatelano

Dira

1. Okeletsa sprite ya tlhware kgotsa o taka sebopego se se kobegileng se na le sefatlhego. Dirisa motheo wa lebokoso la dibuka. Eno ke

kgang ya sekirini, e seng tiro ya go tshela tsela.

2. Tlhopha tlhware. Kgomaretsa boloko jwa Tsamaya-kwa-lefelong ka fa tlase ga Play. Tobetsa pin ya yone ya mmapa mme o baya tlhware gaufi le letlhakore la molema la sethaleng, kgakala le lotlhakore.

3. Okeletsa Tsamaya go ya ka fa ntlheng e e siameng 60 ka lebelo la magareng, morago ga moo bubble ya puo e e reng 'Re fitlhile!' ka metsotswana e le 2. Tlogela sekgala se se lekaneng sa go tsamaya.

Khoutu (recipe)

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TLHWARE MO TSELENG
When Play starts
Go to your chosen left-side start (instant)
Move right 60 (medium)
Say in bubble: We have arrived! for 2 seconds
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Go dira jang

Bala o ya tlase: pele busetsa tlhware kwa tshimologong, morago tsamaya, morago bua. Boloko jwa Tsamaya-kwa-lefelong bo tlhopha felo. Tsamaya e fetola felo foo ka sekgala. Go busetsa go dira gore go tsamaya gongwe le gongwe go simologe mo lefelong le le tshwanang.

Leka

Tobetsa Play gabedi. Tlhware e tshwanetse go simolola mo lefelong le le tshwanang nako nngwe le nngwe. Fetola 60 go nna 30 mme o bapisa. Fa e kopana le lotlhakore, tsamaisa tshimologo go tsena kwa teng kgotsa o kgutshwafatse sekgala.

Fetola

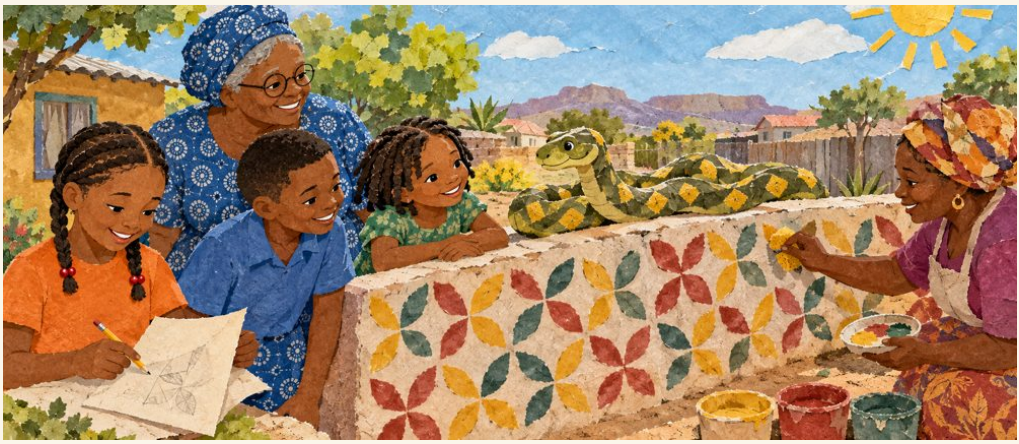
Taka loeto mo pampiring. Akanya sekgala, se leke, morago o se baakanye. Fetola motheo go nna felo mo kgannyeng ya gago, o sa tseye setshwantsho sa aterese.

Tlhalosa le go abelana

Tlhalosa gore ke boloko bofe jo o ka bo tsamaisang fa tlhware e ka bua pele ga go fitlha.

POROJEKE 03

Mokgwa mo lerakeng la ga Nkuku



Kwa ntlheng, Mme Dineo o ne a penta lerako le le kwa tlase la lelwapa la gagwe. O ne a dirisa spongo le mebala e meraro, a dira sebopego se sennye, a se retolola, a se dira gape. Kotara ya potologo nngwe le nngwe. Sebopego sa nna palesa, sa nna naledi, sa nna mokgwa o o neng o bonala o le teng ka dinako tsotlhe.

Lesedi o ne a lebelela a phutheletse le tlhogo mo matsogong. "O itse jang gore a ema leng?" Mme Dineo a tshega. "Fa se boa kwa se simologileng teng." Neo a kopa pensele mme a leka mo pampiring. Letshwao le le lengwe. Retolola letlhare kotara. Letshwao le lengwe. Retolola. Retolola. Morago ga bone, letshwao la gagwe la ntlha le ne le mo emetse kwa godimo.

Kagiso o ne a batla dipotologo di le robedi, nngwe le nngwe e nnye go feta. Lesedi a batla mebala e meraro. Nkuku a tlisa tee mme a lebelela letlhare nako e telele. "Seatla sa gago se a itse seno," a bua. Neo o ne a sa itse gore mokgwa o ka nna bonolo jalo mme o santse o ikutlwala jaaka legae la motho.

Potso ya kgang: *Dipotologo tse kotara di le kae di dira potologo e e feletseng?*

Bokgoni: Go taka le go boeletsa

Dira

1. Okeletsa sprite > Taka. Taka sebopego se sennye jaaka tse di mo lerakeng la ga Mme Dineo, palesa kgotsa naledi, ka sebopego kgotsa borashe. Dirisa Tlatsa fa o batla mmala o o tletseng. Fetsa setshwantsho gore se bonale mo lenaneong la disprite.

2. Tlhopha sebopego sa mokgwa. Ka fa tlase ga Play okeletsa boloko jwa Boeletsa go tswa mo Taolong. Kwala 4.
3. Baya Retolola ka fa molemeng 90 mo teng ga molomo wa Boeletsa. Tlhopha lebelo la magareng gore o bone potologo nngwe le nngwe.

Khoutu (recipe)

```
SEBOPEGO SA MOKGWA
When Play starts
Repeat 4 times:
    Turn clockwise 90 (medium)
```

Go dira jang

Boeletsa e dira dibolokwa tse di mo teng ga yone makgetlo a a balwang. Dipotologo tse nne tsa degere di le 90 di dira degere di le 360. Boloko jwa go retolola bo tshwanetse go nna mo molomong, e seng ka fa tlase kgotsa kwa ntle ga Boeletsa.

Leka

Akanya gore sebopego sa mokgwa se tla lebelela kae morago ga potologo e le nngwe. Tobetsa Play ka Boeletsa 1, morago 2, morago 4. Mofuta wa dipotologo tse nne o tshwanetse go felela o lebile kwa o neng o simologile teng.

Fetola

Penta mokgwa o mongwe mo sebopegong sa gago sa mokgwa. Leka Boeletsa 6 ka dipotologo tsa degere di le 60. Taka mokgwa pele o o dira.

Tlhalosa le go abelana

Letla molekane wa gago gore a supe taelo e e boeletswang. Loop e go boloka go goga boloko bo le bongwe jwa go retolola makgetlo a manè.

PROJEKE 04

Morula e tsaya nako ya yone



Ba ntse ba ya Groot Marico, ba ne ba ema fa tlase ga setlhare sa morula. Fa fatshe go ne go aname ka maungo a a serolwana a a nkgang monate le go bonya. Nkuku a bolelela bana gore setlhare seo se ne se eme foo nako e telele go feta e mongwe a ka e gopolang, le gore ka nako nngwe se ne se sa ntse se se kae go fitlha mo lengwapeng la ga Lesedi. "Ga go na motho yo o neng a nna a lebelela se gola," a bua. "Ke ngwaga e e dirileng jalo."

Kagiso o ne a batla go taka setlhare sotlhe ka nako e le nngwe. A pitlagana thata mme a dira kutu e e jeleng letlhare. Neo a mmontsha mokgwa wa gagwe. A taka setlhare se sennye, morago se se golwane fela mo lakeng la bobedi, morago se se golwane gape. A baya makwalo a mo mothalong. Se sennye. Se segolwane. Se segolwane gape.

"Go tshwana fela jaaka se a gola," Kagiso a bua ka tlhamalalo. Nkuku a dumalana. A re sekirini se ka godisa setlhare mo moweng o le mongwe. Setlhare sa nnete se tlhoka pula le dinako tsa ngwaga le motho yo o sa se gatseleng. Neo a kwala polelo eo ka fa tlase ga setshwantsho sa gagwe, e nnye, gore e se ka ya tseela letlhare sotlhe.

Potso ya kgang: *Diphetogo di le mmalwa tse dinnye di ka bolela kgang e e golang jang?*

Bokgoni: Bogolo le go ema

Dira

1. Taka sprite ya setlhare sa morula se se sa tsofalang, kgotsa o tlhophe emoji ya setlhare jaaka sebopego se se nolofaditsweng

sentle. Se tlhophe.

2. Ka fa tlase ga Play okeletsa Baya bogolo 30 lekgolong. Okeletsa Boeletsa 4.

3. Mo teng ga Boeletsa, baya Ema motsotswana o le 1 morago ga moo Fetola bogolo ka 10. Dirisa lebelo le le bonalang, la magareng.

Khoutu (recipe)

```
MORULA E E SA TSOFALANG
When Play starts
Set size to 30% (instant)
Repeat 4 times:
    Wait 1 second
    Change size by 10
```

Go dira jang

Baya bogolo e fa boleng jwa tshimologo. Fetola bogolo bo okeletsa mo go jone. Go simolola ka 30 le go okeletsa 10 makgetlo a mane go dira 70 lekgolong. Ema e letla yo o lebelelang gore a lemoge kgato nngwe le nngwe. Eno ke setshwantsho sa kgang, e seng tekanyo ya kgolo ya nnete ya setlhare.

Leka

Tobetsa Play gabedi mme o lebelele bogolo. Bo tshwanetse go simolola bo le bonnye mo makgetlong a mabedi. Tlosa go busetsa fela mo porojekeng e e kopilweng mme o bone gore ke ka ntlha yang fa go tshameka makgetlo a mantsi go ka dira gore go tshwenye.

Fetola

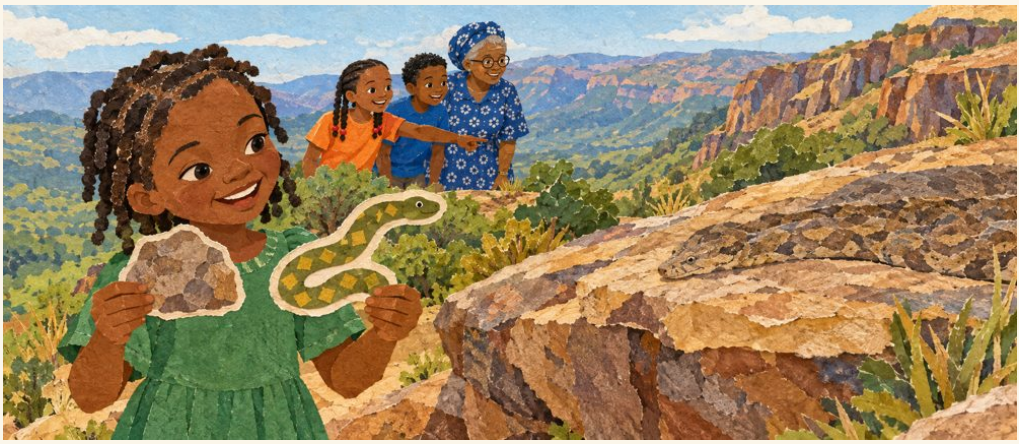
Dira gore morula e e sa tsofalang e fokotsege gape ka go fetola palo go nna -10 mo loopong ya bobedi. Boloka bogolo jwa tshimologo bo le bogolo go lekana gore bo tle bo bonale.

Tlhalosa le go abelana

Dira mothalo wa dipalo wa pampiri wa 30, 40, 50, 60, 70. Bolelela molekane wa gago gore sediriswa se bala eng.

PROJEKE 05

Lefika le le ntseng le panya



Kwa godimo ga tsela ya Magaliesberg, Lesedi a supa lefika le le sephara. "Le le a hema," a bua. Ba bangwe ba tshega. Morago lefika la tshikinyega. Tlhware ya matlapa, e le mmala wa lorole le lemae la maje a a tsofetseng, e ne e robetse mo setlhabelong. E ne e sa ba tsaya kgang sepe. Ba ne ba ema kgakala mme ba tlhokomela gore ba se ka ba tsholetsa mantswe. Nkuku o ne a ba ruta gore o lebelela ka matlho, mme o tsamaya ka dinao.

Kwa gae, Neo a taka dilo tse pedi: lefika, le tlhware ya mmala o o tshwanang. A di sega. A baya tlhware kwa morago ga lefika, morago a sutisa lefika kwa thoko. Tlhware e ne e le teng ka nako yotlhe. Kagiso a ntsha menwana e mebedi jaaka thobane ya moropa. "Tobetsa lefika mme e a bonala."

Ba dira lefika bo konope, tlhware e nna sephiri. Go tsere makgetlo a le mararo pele sephiri se tla ka nako e e siameng. Neo a kwala lefoko le le lengwe kwa morago ga lefika. "Ka bonolo." Ke bontlha jo bo siameng mo go yo o tla le fitlhelang.

Potso ya kgang: *Ke eng se se tshwanetseng go bonala fa moreki wa gago a fitlhela bontlha?*

Bokgoni: Fitlha, bontsha le disprite tse pedi

Dira

1. Okeletsa disprite tse pedi: tlhware e tala ya pampiri e o e pentang le lefika le le sehibidu la bontlha. Di tlogele tsothe mo sethaleng. Baya lefika gaufi le tlhware, e seng godimo ga yone, gore Bontsha e dire gore tlhware e bonale; lefika ke sepheo sa go tobetsa, e seng sekgabo se se sa bonaleng.

2. Tlhopha sprite ya tlhware ya pampiri. Aga setaki sa yone sa Play > Fitlha ka lebelo la ka bonako. Okeletsa setaki se se kgaoganeng sa Tobetsa-simolola mo sprite eno e le nngwe, mme o baye sepheo sa go tobetsa e le sprite ya lefika le le sehibidu.
3. Ka fa tlase ga ketsakgato eo ya go tobetsa okeletsa Bontsha ka 100 lekgolong le bubble e e reng 'Ke e bone!' ka metsotswana e le 2. Tlhware ya pampiri e na le ditiro le fa lefika le le sehibidu le amogela go tobetsa.

Khoutu (recipe)

```
TLHWARE YA PAMPIRI: GO ITLHAMA GA TSHIMOLOGO
When Play starts
Hide (instant)
```

```
TLHWARE YA PAMPIRI: GO BONTSHA
When painted grey rock is tapped
Show at 100% (instant)
Say in bubble: Found it! for 2 seconds
```

Go dira jang

Sprite e le nngwe e ka nna le ditaki di le dintsi. Tlhware ya pampiri e a fitlha fa Play e simolola mme e a bonala morago ga go tobetsa lefika le le sehibidu. Sprite e e fitlhilweng ga e kgone go dirisana, ka jalo o se ka wa tlhoka gore tlhware ya pampiri e e fitlhilweng e le yone e amogele go tobetsa.

Leka

Tobetsa Play, ema gore tlhware ya pampiri e fitlhe, morago o tobetse lefika le le sehibidu. Tlhware ya pampiri e tshwanetse go bonala. Ema mme o tobetse Play gape: e tshwanetse go fitlha gape. Fa lefika le le sehibidu le nyelela, o kwadile khoutu mo sprite e e fosagetseng.

Fetola

Taka bontlha jo bo nang le bokao mo lelapeng la gago. Dira gore bontlha jwa bobedi bo bontshe selo se sengwe se se pentilweng, ka tiro e le nngwe ka nako.

Tlhalosa le go abelana

Tlhalosa gore ke ka ntlha yang fa go fitlha bontlha jo bo le bongwe fela jo bo ka tobediwang go ka dira gore tshwantshetso e se ka ya kgonega.

POROJEKE 06

Nkuku o dumela moranodi



Mo mantsiboeng kwa Groot Marico, bana ba nna mo lorapeng ba reetsa Nkuku a anela kgang. E ne e se kgang e e tswang mo bukeng. E ne e le ya gagwe, ka ga podi e e latlhegileng le ngwanyana yo o botlhale, mme e ne e fetoga go se kae nako le nako fa a e anela. Neo a batla go e boloka. A fitlhela mogala.

Nkuku a emisa seatla. "Kopa pele, ngwanaka." Neo a ikutlwa ditsebe di mo fisa. A kopa. Nkuku a akanya, morago a re, "Ee. Fela ke batla go e utlwa gape." Ba rekota kgang yotlhe, go khutsa pele ga karolo e e tshegisang le tsotlhe. Kagiso a bua ka sephiri kwa bokhutlong ka phoso. Lesedi a tshega thata go fitlha a khotsa.

Fa e boa, Nkuku a reetsa a tswaletse matlho. Morago a kokota tafole. "Ke nna. Mme yo ke mosimane yo o sa kgoneng go didimala." Bana ba dira maekerofouno a pampiri go e kgomaretsa gaufi le setshwantsho. Fa tlase ga yone Neo a kwala: "Kopa pele. Morago rekota."

Potso ya kgang: *Modumo o mokhutshwane ofe o o ka thusang motho go akanya sebaka sa gago?*

Bokgoni: Maekerofouno le modumo

Dira

1. Taka setshwantsho sa Nkuku a anela kgang. Dirisa Okeletsa ka bonako > Okeletsa modumo. Tlhopha Rekota mme, ka tetla ya motho yo o godileng, o rekote modumo wa gago wa pampiri e e tswaletsweng wa metsotswana e 3-5 kgotsa molaetsa o o didimetseng wa go reetsa. O se ka wa rekota mongwe a sa kopa.

2. Mo moeditareng wa modumo, goga dibara tsa kgetho go dikologa karolo e o batlang go e boloka mme o dirise Sega. Reetsa. Rekota gape fa go tlhokega. Fa go se na tetla ya maekerofouno, tlhopha modumo wa lebokoso la dibuka.

3. Tlhopha sprite ya gago. Kgomaretsa Letsa modumo ka fa tlase ga boloko jwa yone jwa Tobetsa-simolola. Tlhopha rekoto ya gago mme o tlhophe 'go fitlha e fela'. Okeletsa bubble morago ga yone.

Khoutu (recipe)

```
SETSHWANTSHO SA MOANEDI
When this sprite is tapped
Play your chosen sound (until done)
Say in bubble: Ask first, then record! for 2 seconds
```

Go dira jang

'Go fitlha e fela' e ema gore modumo o fele pele ga boloko jo bo latelang. 'Mme o tswelele' e letla boloko jo bo latelang go simolola fa modumo o ntse o letsa. Go rekota ke kgetho ya go tlhama; ga go tlhokege lentswe la motho ka namana.

Leka

Tobetsa lekgetlo le le lengwe. Bubble e tshwanetse go tla morago ga fa rekoto e fela. Leka 'mme o tswelele' mo kopong mme o bapise. Lekola bophara jwa modumo le kgetho ya modumo fa go sa utlwale sepe.

Fetola

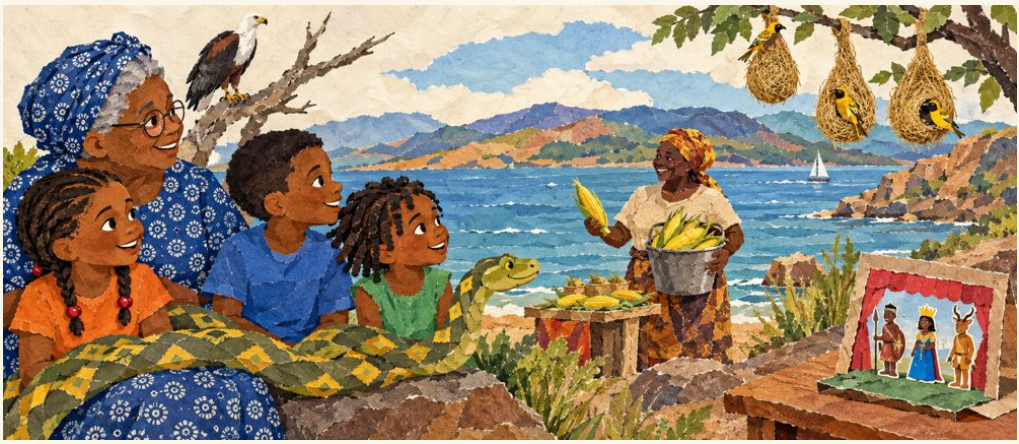
Dira modumo o o sireletsegileng ka pampiri kgotsa diatla. Leka Modumo wa kgakologo mo karolong e e tlhophilweng, morago o bapise le wa tshimologo. O se ka wa goa mo maekerofouneng.

Tlhalosa le go abelana

Letla molekane wa gago gore a tlhalose gore modumo o okeletsa eng se setshwantsho se sa se kgoneng ka bosi. Boloka faele mo sedirisweng ntle le fa motho yo o godileng a letla go e abelana.

PROJEKE 07

Mang o araba eng



Kwa Hartbeespoort Dam, metsi a ne a le a magolo mme a tletse. Nonyane e e jang tlhapi (fish eagle) e ne e bitsa go tswa mo setlhareng se se suleng. Dinonyane tse di dirang dihuta di ne di tlhaba mantswe mo dihuting tsa tsone tse di ritibaditsweng. Mosimane o ne a rekisa mmidi o o monate go tswa mo bokete. Neo a lebelela ka fa sengwe le sengwe se arabelang selo se se farologaneng. Nonyane e e jang tlhapi e arabela tlhapi. Dinonyane tse di dirang dihuta di arabela phefo. Mosimane o arabela batho ba ba feta.

Kwa gae, bana ba dira pontsho ya pampiri mo tafoleng. Kagiso a taka nonyane e e jang tlhapi mme a swetsa gore e tla bitsa fa e tobediwa. Lesedi a taka dinonyane tse di dirang dihuta mme a swetsa gore di tla ritibala fa tafole e šikinyega. Neo a taka tlhware e nnye e thelela mo lebopong, mme a swetsa gore e tla tsamaya fela fa letlhare le retolwa. Ditshwantsho tse tharo. Ditsela tse tharo tsa go tsoga.

"Fa tsotlhe di araba selo se le sengwe, ke lerata fela," Lesedi a bua. Nkuku a baya go roka ga gagwe fa fatshe. "Ke gone o file nngwe le nngwe lekgoro la yone." Ba dira jalo. Pontsho e nnye e ne ya dira jaaka mosong kwa dam, kwa mongwe le mongwe a nang le lebaka la gagwe la go tsamaya.

Potso ya kgang: *Moreki wa gago a ka tshameka mokgwa wa molodi, e seng fela go o lebelela?*

Bokgoni: Ditaki tse di farologaneng tsa ketsakgato

Dira

1. Taka disprite tse tharo tsa ditshwantsho tsa tlhago: nonyane e e jang tlhapi (fish eagle), nonyane e e dirang sehuba le tlhware. Dirisa lebokoso la medumo go okeletsa medumo e meraro e mekhutshwane, kgotsa o rekote go tobetsa go sireletsegileng ka tetla.
2. Tlhopha tshwantsho nngwe le nngwe ya tlhago ka go tshwana. Nngwe le nngwe e fe setaki sa yone sa Tobetsa-simolola le boloko jwa Letsa modumo ka modumo o o tlhophilweng wa tshwantsho eo.
3. Dirisa 'go fitlha e fela' mo setakeng sengwe le sengwe. Rulaganya ditshwantsho gore menwana e di tobetse e se ka ya kopana.

Khoutu (recipe)

```
NONYANE E E JANG TLHAPI
When this sprite is tapped
Play sound A (until done)
```

```
NONYANE YA MOHUTA LE TLHWARE
Make the same stack on each sprite
Choose a different sound for each
```

Go dira jang

Sprite nngwe le nngwe e na le khoutu ya yone. Go ema mo setakeng sa nonyane e e jang tlhapi ga go emise setaki sa nonyane e e dirang sehuba ka tsela ya othomathiki. Diketsakgato tse di kgaoganeng di ka kopana. Mokgwa o o phepa o tswa mo nakong ya motshameki le khoutu.

Leka

Tobetsa nonyane e e jang tlhapi, ema, tobetsa nonyane e e dirang sehuba. Morago leka nonyane e e jang tlhapi le nonyane e e dirang sehuba ka bonako. Reetsa gore a go kopana. Fa tshwantsho nngwe le nngwe e utlwala jaaka e nngwe, lekola lenaneo la modumo la sprite nngwe le nngwe.

Fetola

Taka lebanta la mokgwa: nonyane e e jang tlhapi, nonyane e e jang tlhapi, go didimala, tlhware. Kopa molekane wa gago gore a le tshamekele. Morago o fetole mokgwa o mongwe.

Tlhalosa le go abelana

Neela karata ya mokgwa e e bonalang mo go yo o sa kgoneng go utlwa medumo. Okeletsa kganya e khutshwane mo setakeng sengwe le sengwe sa go tobetsa gore modumo le one o nne le karabo e e bonalang.

PROJEKE 08

Karata ya poso go tswa Taung



Nkuku o ne a ba solofeditse Taung, mme a boloka kano. Ka 1924, a ba bolelela, bagwebi ba kwari ba ne ba bona lehata le lennye mo lefikeng foo. E ne e le la ngwana wa dingwaga di ka nna tharo yo o neng a tshela dimilione tse di ka nnang 2.8 tsa dingwaga tse di fetileng. E ne e le nngwe ya ditshika tsa ntlha tsa batho ba bogologolo tse di fitlhetsweng mo Aforika. Bana ba ne ba eme ba sa tshikinyege mo lefelong leo. Lefelo le ne le ikutlwala le le legolo go feta bone.

Neo o ne a tsere pampiri le pensele, mme a taka thaba jaaka e ne e bonala ka matlho a gagwe. Kwa gae a baya setshwantsho sa gagwe mo tafoleng mme a se tsaya setshwantsho. Tafole yotlhe e ne ya bonala. Kopi e ne ya bonala. Lonao la ga Kagiso lwa bonala. Lesedi a mmontsha gore a sege jang, a taka foreime fa go dikologa fela karolo e e tlhokegang.

Kopi e ne ya nyelela. Lonao la nyelela. Se se setseng e ne e le thaba le loapi lo lo bophara kwa godimo ga yone. "Ngwana," Neo a bua ka bonya. "Ngwana yo o tsofetseng thata, mme re santse re bua ka ene." Nkuku ga a ka a bua sepe, fela a boloka setshwantsho se se segilweng gaufi le bolao jwa gagwe.

Potso ya kgang: *Ke bontlha bofe jwa selo se se tlwaelegileng jo bo tshwanetseng go lebelelwa gaufi?*

Bokgoni: Khamera le go sega ditshwantsho

Dira

1. Ka tetla, Okeletsa sprite > Khamera mme o tseye setshwantsho sa setaki sa thaba se o se dirileng mo pampiring e e sa kwalwang. Tila

difatlhego, dinomore tsa matlo le dilo tsa poraefete. Ditshwantsho di go letla go tlhopha setshwantsho se se leng teng se se amogetsweng.

2. Latela mme o sege go dikologa karolo e o e batlang, kgotsa o dirise sebopego se se bonolo sa go sega. Dirisa phimola go tlosa dikarolo tse o sa di batleng. Setaki sa thaba se se pentilweng ke kgetho e e tshwanang fa khamera e ganetswa.

3. Tlhopha setaki sa thaba. Dira setaki sa go tobetsa se se nang le Retolola ka fa molemeng 20, morago bubble e e tlhalosang bontlha jo bo le bongwe jo o bo lebeletseng.

Khoutu (recipe)

```
SETAKI SA THABA
When this sprite is tapped
Turn clockwise 20 (medium)
Say in bubble: Look at this part! for 3 seconds
```

Go dira jang

Setshwantsho se ka nna sprite, e seng fela motheo. Go sega go fetola karolo ya setshwantsho e o e dirisang. Ga go dire gore tshedimosetso ya poraefete e sireletsege fa e santse e bonala mo setshwantshong.

Leka

Tobetsa lekgetlo le le lengwe mme o lekole gore setaki sa thaba se a retologa le go bua. Tobetsa gape: potologo e nngwe e okeletsa mo go ya ntlha. Seno se dirilwe ka maikaelelo. Ema mme o se busetse morago fa o batla thulaganyo ya tshimologo.

Fetola

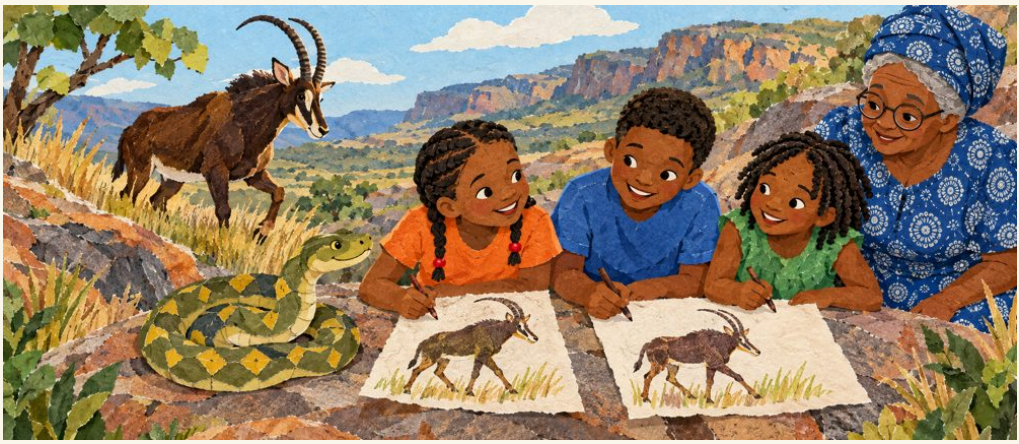
Penta setaki se se akanyeditsweng sa thaba gaufi le setshwantsho. Letla nngwe le nngwe go tlhalosa bontlha jo bo farologaneng. Bua ka go lebelela le go itlhama.

Tlhalosa le go abelana

Pele ga go romela sepe, tlhatlhoba setshwantsho sotlhe mme o reetse modumo mongwe le mongwe go batla tshedimosetso ya poraefete. Motho yo o godileng o dira tshwetso ya bofelo ya go abelana.

PROJEKE 09

Kgaswane e e tsamayang



Kwa Kgaswane Mountain Reserve kwa godimo ga Rustenburg, Nkuku a tlhalosa gore tshireletso eno e rotilwe ka kgaswane (sable). Mothusi wa tsela a ba boleletse gore phologolo e ntsho e e nang le dinaka tse di telele tse di kobegileng e nna mo dithabeng tseo, mme bana ba lebelela thaba nako e telele pele ga ge e nngwe e tswa. Sable e ne ya tsholetsa leoto la kwa pele, morago le lengwe. E ne e sa potlakele.

Lesedi a leka go e taka e tsamaya, mme a retelelwa. Setshwantsho sengwe le sengwe se ne se tshwana le seemo. Neo a akgela ditshwantsho tse pedi: se le sengwe se le leoto la kwa pele le le fa pele, se sengwe le le kwa morago. Kagiso a di fetola gantsi. Sable ya tsamaya. E ne e le go tsamaya go go sa tlhwatlhwa, go tshegisang, go ka nnang go fitlhetse, mme e ne e le go tsamaya.

"Go dira jang?" Kagiso a botsa. Nkuku, yo o neng a sa ke a fa karabo e khutshwane fa e telele e ka dira, a re leitlho le tlatsa se se tlhaelang. Motsamaisi yo o siameng o tlhoka ditshwantsho tse pedi fela le motho yo o lebelelang. Bana ba ne ba tshwere pampiri tse pedi mo diatleng tsa bone jaaka dineo tse dinnye.

Potso ya kgang: *Ke karolo efe e e tshwanetseng go fetoga, mme ke efe e e tshwanetseng go nna e ntse e eme?*

Bokgoni: Diaparo le motsamao o o bonalang

Dira

1. Dirisa OctoStudio 1.1.0 kgotsa e ntšhwa. Penta sprite ya sable. E baakanye go tswa mo lenaneong la disprite go bula diaparo tsa yone mo Moeditareng wa Pente.

2. Kopa kgetlo la ntlha la seaparo. Fetola fela maemo a leoto la kwa pele mo la bobedi. Letlalo la onione le ka bontsha setshwantsho sa pele gore mmele o nne o rulagantswe sentle.
3. Tlhopha sable. Ka fa tlase ga Play, okeletsa Boeletsa 4 le Letsa diaparo mo teng. Baya gore e letse diaparo tsa gago tse pedi tse di bonalang ka lebelo la magareng.

Khoutu (recipe)

```
SABLE
When Play starts
Repeat 4 times:
    Play costumes: 2 (medium)
```

Go dira jang

Diaparo ke ditebego tse di farologaneng tsa sprite e le nngwe. Boloko jwa Letsa diaparo bo di bontsha ka tatelano. Maemo a mabedi a maoto a boeleditswe makgetlo a mane a dira motsamao o o bonalang o o bonolo. Seno se farologane le go tsamaisa sprite yotlhe mo sethaleng.

Leka

Tobetsa Play mme o lebelele fela mmele. Fa o tlolela, rulaganya ditshwantsho mo Moeditareng wa Pente. Fa diaparo di tlhaela, lekola kgatiso ya app le gore diaparo tsothe di a akaretswa.

Fetola

Okeletsa seaparo sa boraro gareng ga leoto le le fa pele le le kwa morago. Bapisa motsamao. Kgotsa o tlhophe emoji e e nang le diaparo di le dintsi go na le go taka.

Tlhalosa le go abelana

Bontsha ditshwantsho tse pedi tse di eme pele ga o bontsha motsamao o o bonalang. Letla molekane wa gago gore a tlhalose gore ditshwantsho tse di eme di nna motsamao jang.

PROJEKE 10

Dikganya tse tharo fa godimo ga lekgwa



Pilanesberg e bonala jaaka mokoloko wa dithaba o o nang le sekgwa se segolo gare. Dithaba tseno di tsofetse thata, di tsofetse go feta sengwe le sengwe se bana ba ka se akanyang. Nkuku a ba anela kgang ka bonnete. Ga se thaba ya mollo e e tlhaselang. Ke se se setseng sa selo se se neng sa tonya kwa tlase ga lefatshe, morago sa nyelela ka bonya go fitlha mokoloko o setse fela. Neo a bua ka tsebe "dingwaga di le bilione" mme a se kitla a itse gore a re eng gape.

Ba lebelela dikganya tse di farologaneng tse tharo go tswa mo thabeng e le nngwe. Ka moso thata, ka moswe mo sekgweng. Motshegare, fa bogobe jwa bophelo bo ne bo nna mmala wa borotho. Ka maitseboa, fa dithaba di ne di nna mmala wa tee. Kwa gae, Lesedi a penta nngwe le nngwe e le motheo mo letlharing la yone, mme Kagiso a di kgomaretsa mo mothalong.

Fa Neo a fetola letlhare, tlhware e ne e sa fetoge. Ke lefatshe fela le le kwa morago ga yone le le fetogileng. Ke jalo nako e neng e ikutlwala ka teng mo dithabeng, a swetsa jalo. Phologolo e le nngwe, magodimo a mararo, le mokoloko wa lefika o o sa tlhokeng go potlaka.

Potso ya kgang: *Moreki wa gago o tla itse jang gore nako e fetogile?*

Bokgoni: Metheo le ditiragalo

Dira

1. Okeletsa metheo e mebedi ya lebokoso la dibuka mme o e rulaganye mo dintlheng tsa gago e le motshegare le maitseboa. Dirisa mafelo a mabedi afe fela a a farologaneng sentle; ga a tlhoke maina a a lebisitsweng mo app. Motheo wa boraro wa mo mosong ke kgetho.
2. Penta sprite ya moanedi o o akanyeditsweng wa tlhware. Setaki sa yone sa Play se fetolela kwa motheong wa gago wa ntlha ka bonako, se ema metsotswana e le 3, mme se fetolela kwa wa bobedi.
3. Mo moanedi yo o tshwanang, okeletsa setaki se se kgaoganeng sa tshimologo ya motheo sa motheo wa maitseboa, se latelwa ke bubble ya puo.

Khoutu (recipe)

```
MOANEDI: MOTHALO WA NAKO
When Play starts
Switch backdrop to first scene (instant)
Wait 3 seconds
Switch backdrop to evening scene
```

```
MOANEDI: KARABO YA MAITSEBOA
When evening backdrop starts
Say in bubble: Same hills, new light. for 3 seconds
```

Go dira jang

Motheo o ka simolola setaki se sengwe. Kaedi ya semmuso e re ketsakgato eno e simologa fa gare ga phetogelo. Tlhopha diphetogelo tsa ka bonako fa lefelo la gago le tlhoka phetogo e e bonalang gone jaanong, mme o leke nako.

Leka

Tobetsa Play gabedi. Lefelo la ntlha le tshwanetse go boa pele ga phetogo ya maitseboa. Fa karabo e sa tle, tlhomamisa gore ketsakgato le boloko jwa go fetola di tlhopha motheo o le mongwe wa bobedi.

Fetola

Okeletsa modumo o le mongwe wa lefelo lengwe le lengwe. Rekota polelo ya gago e e didimetseng ya maitseboa ka tetla, kgotsa o tlogele kgang e le ya go lebelelwa fela.

Tlhalosa le go abelana

Anela kgang ka ditshwantsho tse pedi tsa pampiri pele. Morago o tlhalose gore ketsakgato efe e tsamaisa kgang ya gago ya dijithale kwa motsotsong o o latelang.

POROJEKE 11

Botlhe ba a nna, botlhe ba a bua



Ka Sontaga, lelapa le ne le nna mo lelwapeng mo mokoloko o o sa rulaganyang wa ditulo. Go ne go na le tshwetso e e tshwanetseng go dirwa ka gore ke mang yo o tla isang Nkuku kwa kokelong ka Labobedi. Mongwe le mongwe o ne a na le kakanyo. Kagiso a simolola pele rremogolo a ntse a sa fetsa. Nkuku a tsholetsa monwana o le mongwe. "Ka tsela ya bogologolo," a bua, "kgotla ke lefelo kwa batho ba nnang teng mme ba bua ka tatelano. Yo o tshwereng thobane ke yo o buang."

Ba bona thobane. E ne e tsamaya go tswa mo seatleng go ya kwa seatleng. Mang kapa mang yo o e tshwereng o ne a bua, ba bangwe ba reetsa. E ne e le bonya. Ke gone Neo a lemoga gore ke lekgetlo la ntlha a utlwa polelo yotlhe ya malome.

Morago bana ba e dira motshameko o monnye mo pampiring. Tlhware e e takilweng e fetisetsa molaetsa kwa nonyaneng e e takilweng, nonyane e e fetisetsa kwa podi e e takilweng, jalo jalo. Ga go na ope yo o buang go fitlha molaetsa o tla. Lesedi a okeletsa sebopego sa bofelo se se reng fela, "Ke a leboga." Go ne go ikutlwala e le tsela e e siameng ya go felela.

Potso ya kgang: *Khoutu e ka fetisa nako ya go bua jang ntle le gore mantswe a mabedi a kopane?*

Bokgoni: Melaetsa le go tsaya dikhutlo

Dira

1. Okeletsa disprite tse pedi tsa babui: thobane ya go bua le motheetsi wa tlhware. Okeletsa modumo o mokhutshwane wa lebokoso la

dibuka mo go nngwe le nngwe, kgotsa o rekote mela e mebedi e mekhutshwane ya gago ka tetla.

2. Mo go mmui A, letsa modumo wa ntlha 'go fitlha o fela', morago Romela molaetsa o dirisa lenaneo la melaetsa. Tlhopha molaetsa o o leng teng mme o kwale kgetho ya one ka tlhamalalo.

3. Mo go mmui B, okeletsa Fa molaetsa o amogelwa ka kgetho e e tshwanang, morago o letse modumo wa bobedi 'go fitlha o fela'. Eno ke molaetsa wa mo porojekeng, e seng molaetsa o o romelwang kwa go motho.

Khoutu (recipe)

```
MOANEDI A
When Play starts
Play first line (until done)
Send chosen message to all
```

```
MOANEDI B
When chosen message is received
Play second line (until done)
```

Go dira jang

Romela e neelana ka matshwao. Boloko jwa go amogela ke boloko jwa go simolola. Ke molaetsa o o tshwanang fela o o simololang setaki sa B. Modumo wa ntlha o fela pele ga letshwao le romelwa, ka jalo mantswe a a rulagantsweng a tsaya dikhutlo.

Leka

Tobetsa Play lekgetlo le le lengwe mme o reetse. A e tshwanetse go fela pele ga B e simolola. Fa B e didimetse, bapisa dikgetho tsa melaetsa. Fa B e simologa ka bonako, lekola gore setaki sa yone se simolola ka Amogela, e seng Play.

Fetola

Letla B gore e romele molaetsa o mongwe wa moanedi wa boraro. Dirisa le dikheshene gore bokao bo nne teng ntle le go utlwa.

Tlhalosa le go abelana

Tlhalosa phapang gareng ga letshwao leno la khoutu la selegae le go romela faele ya porojeke kwa go mongwe.

POROJEKE 12

Thaba e fetoga ditlhase tsa mollo



Mo ngwaneng wa maitseboa, Magaliesberg ya fetoga. Mmala o o sunyang wa lefika wa nna mogote, morago wa nna oranji, morago wa nna pinki e bana ba neng ba se na lefoko ka yone. Neo a tshwara moya. Kagiso a re e tshwana le melora fa Nkuku a tsosa molelo. Lesedi a re e tshwana le teng ga lehapu.

Ba leka go e penta mme ba retelelwa ka ditsela di le tharo tse di farologaneng. Mmala o o sunyang o ne o le sephara thata. Oranji e ne e le lerata thata. Nkuku a akgela go simolola ka mmala o o lefifi mme a okeletsa kganya e nnye kwa godimo, gape le gape. Ba dira jalo. Ka bonya pampiri ya simolola go phatsima.

"Thaba e le nngwe, mantsiboa a le mongwe," Nkuku a bua, "mme mmala ga o ke o tshwana gabedi." Neo a dira tlhware ya gagwe e le tala e e sa lebegeng thata e e nang le mmaba o mosesane wa gouta, jaaka e kete tlhaba ya letsatsi e e ntshitse go e ama. A e tsholetsa gaufi le thaba mme, ka motsotswana, ya bonala e le ya fa.

Potso ya kgang: *Maikutlo afe a o ka a tlhamang ka mmala go na le ka mafoko?*

Bokgoni: Kganya le mmala

Dira

1. Penta sprite ya tlhware mo motheong o o lefifi wa lebokoso la dibuka, jaaka thaba morago ga letsatsi le phirima. E tlhophe.

2. Kgomaretsa Phatsima ka mmala o o mogote ka fa tlase ga boloko jwa Tobetsa-simolola. Morago kgomaretsa Baya mmala wa sprite, o tlhopha mmala o o o ratang.

3. Fetola dikgetho tsa mmala mme o bapise. O se ka wa dirisa boloko jwa lenaneo la tshipi la mmele mo porojekeng eno.

Khoutu (recipe)

```
TLHWARE E E PHATSIMANG
When this sprite is tapped
Glow with your chosen warm colour
Set sprite colour to your chosen colour
```

Go dira jang

Phatsima e dira mogopo wa lesedi go dikologa sprite. Baya mmala o fetola mmala wa yone. Tseno ke ditlamorago tsa sekirini. Ga go tlhokege lebone la nnete kgotsa lesedi le le panyang. Go tlosa mmala o o beilweng, tlhopha ga go na mmala mo lenaneong la boloko joo.

Leka

Tobetsa lekgetlo le le lengwe mme o bapise tlhware pele le morago. Fa dintlha tsa yone di nyelela, fetola mmala kgotsa o nolofatse setshwantsho. Tila go panya ka bonako kgotsa diphetogo tse di potlakileng tsa mmala.

Fetola

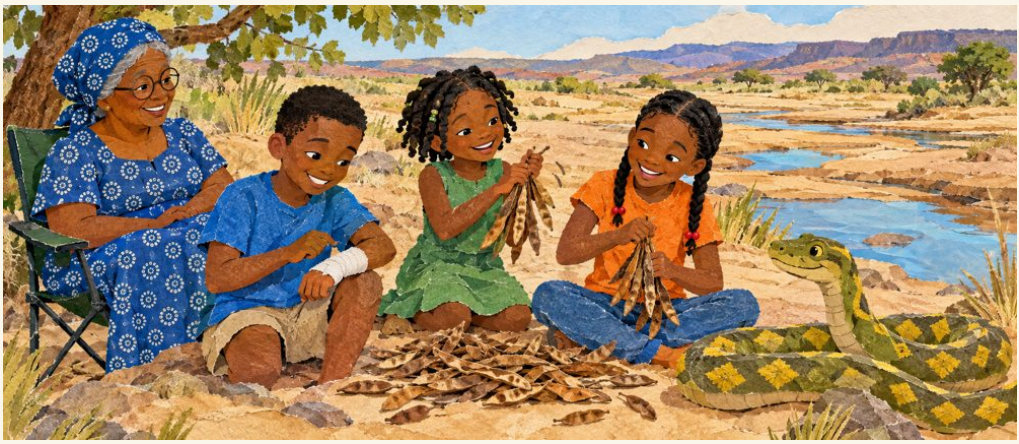
Dira lefelo la bobedi le le nang le mmala o o tsididi o o ikhutsang. Botsa molekane wa gago gore o akanyetsa maikutlo afe, o sa mmolelele karabo ya gago pele.

Tlhalosa le go abelana

Neela mofuta o o sa phatsimeng mo go mongwe le mongwe yo o sa ikutlweng sentle ka ditlamorago tsa lesedi. Ga go na ngwana yo o tlhokang go lebelela sekirini se se panyang go ithuta mebala.

POROJEKE 13

Seatla se se sa kgoneng go šikinyega



Kwa Molopo, mokero o ne o omile gantsi, o na le matamo a metsi mo noka e gopolang ka teng. Bana ba ne ba nna mo lebopong ba na le kgetsana ya dikgogwana tse di omileng. Kagiso a šikinya kgogwana ya gagwe mme ya tlhaba. Lesedi a šikinya ya gagwe. Neo a šikinya ya gagwe. Mongwe le mongwe a dira tladi e nnye e e omileng.

Morago Kagiso a ntsha sefatlhego. O ne a itsholetse mo seatleng bosigo jo bo fetileng mme a sa kgone go šikinya sepe. A leka go e fitlha, a tsenya seatla fa tlase ga leoto. Lesedi a lemoga. A baya kgogwana ya gagwe fa fatshe. "Mo motshamekong wa rona, go šikinya go dira gore tlhware e tsamaye. Ke eng fa go tobetsa le gone go dira jalo?"

Ba fetola leano. Go šikinya kgotsa go tobetsa, nngwe fela e dira gore tlhware ya pampiri e thelele. Seatla sa ga Kagiso sa tlogela go tlhoka. Neo a re ka bonya motshameko ga o a tshwanela go nna teko ya gore mmele wa mang o dira botoka. Nkuku, a reetsa a le mo setulong sa gagwe se se kobegang, a re fela, "Go siame. Jaanong botlhe ba mo motshamekong."

Potso ya kgang: *Porojeke e le nngwe e ka amogela ditsela tse di farologaneng tsa go tsamaya jang?*

Bokgoni: Šikinyega, le tsela e nngwe ya go tobetsa

Dira

1. Tlhopha sprite ya tlhware ya pampiri. Dira setaki sa Šikinya-simolola se se nang le Tlola 20 le bubble e khutshwane. Simolola ka kutlwisiso

e nnye ya go šikinya.

2. Okeletsa setaki sa bobedi mo sprite e e tshwanang se simologang fa e tobediwa. Se fe go tlola le bubble e e tshwanang gore go šikinya e nne kgetho.

3. Tshwara mogala kgotsa tablet ka diatla tsa mabedi fa godimo ga selo se se bonolo. Dira motsamao o o bonolo fela. Mo ho Chromebooks, dirisa konope ya go šikinya ya sekirini.

Khoutu (recipe)

```
TLHWARE YA PAMPIRI: TSELA YA GO ŠIKINYA
When device is gently shaken
Jump 20
Say in bubble: There is a place for you! for 2 seconds
```

```
TLHWARE YA PAMPIRI: TSELA YA GO TOBETSA
When this sprite is tapped
Jump 20
Say in bubble: There is a place for you! for 2 seconds
```

Go dira jang

Ketsakgato ya sensara e golaganya tiro ya mmele le khoutu. Ketsakgato ya go tobetsa e ka naya dipholo tse di tshwanang. Dilo tse di tlhokegang tsa hardware le motsamao di a farologana, ka jalo tsela e nngwe e e kgonegang ke karolo ya tlhamo, e seng porojeke e e kwa tlase.

Leka

Leka go šikinya ka bonolo lekgetlo le le lengwe, morago o tobetse lekgetlo le le lengwe. Nngwe le nngwe e tshwanetse go dira karabo e le nngwe e e bonalang. Fa go šikinya go ipoeletsa ka bonako, baakanya kutlwisiso ya yone kgotsa o dirise tsela ya go tobetsa fela.

Fetola

Dirisa diaparo tse pedi go na le go tlola fa o nang le tsone. Boloka motsamao o le bonya go lekana gore o ka o latela.

Tlhalosa le go abelana

Botsa molekane wa gago gore o rata tsela efe. O se ka wa mo gapeletsa go šikinya kgotsa go tshwara sediriswa ka tsela e e sa ikutlweng sentle.

PROJEKE 14

Mokoloko wa dithaba tsa bogologolo mo fatshe ga kicheng



Pula e ne ya ba tshwara mo teng. Nkuku a kgotlolola lefelo mo fatshe ga kicheng mme Kagiso a simolola go dira Pilanesberg ka khatene, fela jaaka ba e bone, mokoloko wa dithaba tse dinnye le sekgwa gare. E ne e se sekao se se itekanetseng. E ne e le sekao se se reng, se ke sebopego sa sengwe se se tsofetseng thata.

Neo a taka tlhware e nnye mme a e baya kwa lotlhakoreng. Lesedi a taka leralala la metsi gare. Tlhware e ne e tlhoka go fitlha kwa metsing, fela e se ka go kgotlhana le dithaba tsa khatene. Neo a leka go e tsamaisa ka go sega boto. Go tshwana thata, tlhware ya kgotlhana. Go nnye thata, ga go na se se tsamayang.

Mo bokhutlong a leka go sega go gonnye thata, morago go gongwe. Tlhware ya thelela kwa metsing mme ya ema fa a ama metsi. Neo a ikhutsa morago. O ne a ithutile sengwe se a neng a sa se solofela: tshimologo e bonya e dira bokhutlo jo bo botoka. Kagiso a kokota fa fatshe jaaka moropa gore a dumele.

Potso ya kgang: *Go direga eng fa o tsamaisa ka dipotologo tse dinnye?*

Bokgoni: Go sega le ketsakgato ya go ama

Dira

1. Penta sprite ya tlhware le sprite e kgolo e e pentilweng ya sepheo sa leralala la metsi. Baya sepheo kgakala le tlhware ya tshimologo gore di se ka tsa ama. Dira setaki sa Play mo tlhwareng se se yang kwa

felong la tshimologo le le tlhophilweng.

2. Morago ga go busetsa, dirisa Sega go tswa mo Dibolokong tse dingwe ka fa teng ga Ka metlha. Gantsi e tla le boloko jwa Ka metlha. Baya ditsela tsotlhe tse nne le lebelo le le bonya. Tshwara sediriswa ka kgono ya sone e e ikgethileng pele ga Play.

3. Mo tlhwareng, okeletsa ketsakgato ya go simolola go ama e e lebisitsweng kwa sepheong. Okeletsa bubble ya moketeko morago ga moo Emisa ditaki tsotlhe. Mo Chromebook dirisa kgatiso ya yone ya go sega ya sekirini.

Khoutu (recipe)

```
TLHWARE: GO TSAMAISA
When Play starts
Go to chosen start position (instant)
Forever:
    Tilt to move (four directions, slow)
```

```
TLHWARE: GO FITLHA
When python touches goal
Say in bubble: Water! for 2 seconds
Stop all scripts
```

Go dira jang

Ka metlha e tswelela go lekola go sega fa motshameko o dira. Ketsakgato ya go ama e tlhoka disprite tse pedi. Emisa tsotlhe e khutlisa ditaki tse di dirang tsa porojeke; konope ya Ema le yone e ka e emisa nako nngwe le nngwe.

Leka

Boloka disprite tsotlhe di le kgakala le dikhutlo. Tsamaisa ka bonya kwa sepheong. Lekola gore motshameko o a ema. Tshameka gape go busetsa tlhware. Fa sediriswa se se na le go sega go go dirisegang, dirisa taolo tsa Chromebook kgotsa tsela e nngwe e e fa tlase.

Fetola

Tsela e nngwe: Emisa karolo ya Ka metlha/go sega ka disprite tse nne tsa metshwari. Mo tlhwareng dira ditaki tse nne tsa sepheo sa go tobetsa, nngwe le nngwe e e e tsamaisang dikarolo di le 10 mo ntlheng ya metshwari eo.

Tlhalosa le go abelana

Taka tsela pele ga go tshameka. Dira tsela e e bophara go na le sekgwa se se thata. O se ka wa sega Chromebook ka mmele.

POROJEKE 15

Dikgakana tse tharo kwa morago ga tanka



Kgaka e ne e fitlha sehuba sa yone kwa morago ga tanka ya metsi. Ka Labone, dikgakana tse tharo tse di nang le mebala e e tlhakaneng di ne tsa latela mmaagwe go tswa mo lelwapeng la ga Nkuku. Legodimo le ne le lefifile le le boima. Tladi e ne e rotha kwa nngwe kwa godimo ga Mahikeng.

"Di bone pele ga pula," Nkuku a bua. Dikgakana di ne di le dinnye le go potlaka. E nngwe e ne e le fa lefikeng. E nngwe e ne e le gaufi le bokete. E nngwe e ne e tshologetse kwa morago ga tanka gape.

Neo a dira motshameko mo pampiring pele. Dikgakana tse tharo tse di takilweng. Nako nngwe le nngwe fa a bona nngwe, a e tobetsa, mme ya tlola mo sehubeng se sennye se a se takileng kwa khutlong. Lesedi a batla lebokoso le le bolokang palo. Go tobetsa gongwe le gongwe go okeletsa o le mongwe.

Fa ditlhwatlhwa tsa ntlha tsa pula di wela, lebokoso le ne la re tharo. "Tharo di bolokegile," Kagiso a bua. Nkuku a tsena kwa teng go batla bokete jwa go kgoga lerumo la marulelo. Neo a kwala palo mo seatleng sa gagwe, gore a se ka a e lebala.

Potso ya kgang: *Palo e ka gopola tiro jang ntle le go e bala gabedi?*

Bokgoni: Dibolokwa tse di fetogang le dintlha

Dira

1. Penta disprite tse tharo tse di bonalang tsa dikgakana mo motheong wa lelwapa. Dirisa ditshwantsho, e seng diphologolo tsa nnete. Mo

Dibolokong tse dingwe, bula lenaneo la bolokwa se se fetogang mme o dirise + go dira 'dikgakana'. Maina a dibolokwa a ka se tlose gompieno, ka jalo tlhopha ka kelotlhoko.

2. Mo kgakeng A, aga Play > Baya dikgakana go 0 > Bontsha ka 100%. Mo dikgakaneng B le C dirisa Play > Bontsha ka 100%. Dibolokwa di bontshiwa ka tlwaelo; dirisa Bontsha bolokwa se se fetogang go tsha fa go tlhokega.

3. Mo kgakaneng nngwe le nngwe, okeletsa Tobetsa sprite eno > Fetola dikgakana ka 1 > Fitlha ka bonako. Fa e fitlhilwe, ga e kake ya tobediwa gape. Tobetsa Play mme o ema gore tshimologo e fele pele ga go kgobokanya.

Khoutu (recipe)

```
KGAKANA A: TSHIMOLOGO
When Play starts
Set variable chicks to 0
Show at 100% (instant)
```

```
KGAKANA NNGWE LE NNGWE: GO BALA
When this sprite is tapped
Change variable chicks by 1
Hide (instant)
```

Go dira jang

Bolokwa se se fetogang ke palo e e nang le leina e khoutu ya gago e ka e ntšhafatsang. Baya e dira lefela le le ntšhwa. Fetola e okeletsa le le lengwe. Fitlha e thibela gore kgakana e le nngwe e se ka ya kgobokanngwa gape. B le C di tlhoka ditaki tsa tsone tsa Bontsha-mo-Play, e seng go busetsa palo gape.

Leka

Morago ga Play, tobetsa kgakana nngwe le nngwe lekgetlo le le lengwe. Solofela 1, 2, 3. Tobetsa felo go go senang sepe fa kgakana e neng e le teng: ga e a tshwanela go okeletsa gape. Simolola gape mme o solofele 0 le dikgakana tsotlhe di bonala. Tobetsa ka bonya fa o tlhatlhoba.

Fetola

Dira dikgakana di le tlhano mme o fetole sepheo. Boloka mongwe yo o busetsang a le nosi gore go nne bonolo go mmona. Leka go tobetsa ka bonako mo sedirisweng sa nnete pele o se dirisa jaaka motshameko wa phaposi ya borutelo.

Tlhalosa le go abelana

Bolelela molekane wa gago gore ke ka ntlha yang fa 'fetola dikgakana ka 0' e ka se busetseng dintlha tsa maabane.

PROJEKE 16

A botlhe ba tharo ba gae?



Ka moso o o latelang pula e ne e sa tlhole e le teng mme lelwapa le ne le nkga mmu o o kolobileng. Kgaka e e tshadi e ne e tsamaya fa pele ga dikgakana tsa yone jaaka lepodisa le lennye le le ikgantshang. Nkuku a nna mo setepisong a na le tee ya gagwe. "A le a itse gore botlhe ba teng?" a botsa.

Kagiso a bala. Morago Lesedi a bala. Ba file dikarabo di le pedi tse di farologaneng, mme kgaka e ba lebelela ka bobedi jaaka e kete ba tlhoka go potlaka.

Neo a bula porojeke e a neng a e dirile ka maabane. A okeletsa kgaka jaaka sebopego se sešwa. Morago a e fa molao. Fa kgaka e tobediwa, e lekola palo. Fa e feta bobedi, e re, "Ke a leboga! All three are home." Fa go sa nna jalo, e re, "Find all three chicks first."

Neo a tobetsa kgaka morago ga kgakana e le nngwe. Morago ga tse pedi. Morago ga tse tharo. Kgaka e ne ya leboga fela lekgetlo la bofelo. "O thata go feta wena, Nkuku," Lesedi a bua.

Potso ya kgang: *Porojeke ya gago e tshwanetse go dira eng fa sepheo se sa fitlhelwa?*

Bokgoni: Ditlhokego le sepheo

Dira

1. Kopa Porojeke 15. Okeletsa sprite ya kgaka, o boloke khoutu ya yone e kgaogane le ya dikgakana.
2. Mo kgakeng, kgomaretsa boloko jwa Fa ka fa tlase ga Tobetsa sprite eno. Tsenya morepoti wa bolokwa se se fetogang sa dikgakana mo lekgobong la ntlha la palo. Tlhopha go feta mme o baye 2 mo lekgobong le lengwe.

3. Mo teng ga Fa baya bubble 'Ke a leboga! All three are home.' ka metsotswana e le 2. Tobetsa tlhakelo ya boloko jwa Fa go okeletsa Go sa nna jalo; mo teng ga Go sa nna jalo baya 'Find all three chicks first.' ka metsotswana e le 2.

Khoutu (recipe)

```
KGAKA
When this sprite is tapped
If chicks > 2:
    Say: Ke a leboga! All three are home. for 2 seconds
Else:
    Say: Find all three chicks first. for 2 seconds
```

Go dira jang

Fa e lekola maemo lekgetlo le le lengwe, fa go tobetsa go e fitlha. Ka ntlha e le nngwe ya kgakana nngwe le nngwe, dikgakana di feta 2 di kaya gore tse tharo tsotlhe di kgobokeditswe. Go sa nna jalo ke tsela e nngwe. Tlhatlhobo eno e e tsamaiswang ke ketsakgato ga e tlhoke Ka metlha ka gonne go tobetsa go gontsha nngwe le nngwe go dira tlhatlhobo e ntšhwa.

Leka

Tobetsa kgaka fa dikgakana di le 0, 1, 2 le 3. Ke ya bofelo fela e tshwanetseng go naya molaetsa wa tebogo. Simolola gape mme o lekole gore kgaka e naya molaetsa wa go se fele gape.

Fetola

Fa go na le dikgakana di le tlhano, fetola maemo go nna dikgakana > 4. Bolela gore ke ka ntlha yang fa go fetola setshwantsho fela go ka tlogela molao wa kgale o le phoso.

Tlhalosa le go abelana

Dira setshwantsho sa pampiri sa ditsela tse pedi: sepheo se fitlhetswe / ga go ise. Fa maikutlo a a thusang mo ditseleng tsotlhe tse pedi.

PAKETE 17

Ledimo le le tlileng go tswa kae



Fa tlase ga Pilanesberg, Nkuku a anela kgang. "Ba re," a simolola, e leng ka fa a neng a anela tsa bogologolo ka teng, "gore mo dikgannyeng dingwe tsa Bakgatla go na le noga e kgolo e e bidiwang Kwanyape. Go twe e tsamaya e le leloba le le tshesane. Ga o ke o itse gore e tla nna kgolo go le kae, kgotsa e tla tswa kae."

Neo a lebelela lorole mo tseleng. Leloba le lennye la tsogela motsotswana mme la nyelela. Bana ba tshega mme ba tlogela go tshega, fa go ka diragala.

Kwa gae, Kagiso a re, "A re dire leloba le le nang le bogolo bo bo farologaneng nako nngwe le nngwe." Lesedi a ntsha dae. Neo a sega leloba la pampiri mme a taka dikgolo tse thataro, go tswa mo go nnye go ya kwa go golo. Dae e ne e tlhopha bogolo. Ka dinako tse dingwe e ne e le sefokotsana. Lekgetlo le le lengwe e ne e batla e tletse letlhare lotlhe. Ka dinako tse dingwe go latlha gabedi go ne go naya bogolo jo bo tshwanang.

Nkuku a re e ne e le kgang fela, mme kgang e e siameng ga e tlhoke go dumelwa gore e ratwe. Bana ba boloka leloba gare ga letlhare mme ba letla dae gore e tlhophe.

Potso ya kgang: *A tlhopho e e sa rulagangwang e ka boeletsa, mme e santse e sa rulagangwa?*

Bokgoni: Dipalo tse di sa rulagangwang

Dira

1. Okeletsa sprite ya ledimo, sefifi sa lorole se o se pentang. Mo ketsakgatong ya go tobetsa kgomaretsa Baya bogolo. Mo lekgobong la yone la palo, gokelela morepoti wa Palo e e sa rulagangwang go tswa mo Dibolokong tse dingwe.
2. Baya bophara jwa go sa rulaganyeng go tswa kwa 40 go fitlha kwa 100. Tlhopha diphetogo tsa bogolo tsa ka bonako gore pholo e bapisiwe ka bonolo.
3. Boloka ledimo gaufi le bogare jwa sethale. Dipalo tse dikgolo tsa bogolo di ka tlhoka setshwantsho sa ntlha se se nnye.

Khoutu (recipe)

```
LEDIMO
When this sprite is tapped
Set size to random number from 40 to 100 (%)
```

Go dira jang

Morepoti o o kgolokwang o naya palo kwa go boloko jo bongwe. Ga o nne o le nosi jaaka tiro. Bophara bo baya ditekanyo tse di kgonegang. Go tobetsa gabedi go ka naya boleng jo bo tshwanang; go sa rulagangwe ga go reye gore ka metlha go farologane.

Leka

Tobetsa makgetlo a le matlhano mme o bapise dikgolo. Nngwe le nngwe e tshwanetse go nna fa gare ga dintlha tsa gago tse o di tlhophileng. Fa go sa fetoge sepe, lekola gore morepoti o kgomaretswe mo teng ga lekgoba la bogolo, e seng a robetse a lokologile gaufi.

Fetola

Leka 50-60 mme o bapise le 40-100. Ke bophara bofe jo bo dirang gore diphetogo di nne bonolo go di lemoga? Kwala tebelelopele pele.

Tlhalosa le go abelana

Golaganya motshameko wa dae wa nnete le tlhopho ya palo ya sekirini. Ka bobedi di tlhoka sethaopo se se tlhamaletseng sa diphelelo tse di kgonegang.

POROJEKE 18

Lefoko la mokoloko ka bone



Nkuku a ba tseela kwa Madikwe Game Reserve mme a ba bolelela mo tseleng gore ka 1991, pholoso e kgolo e e bidiwang Operation Phoenix e ne ya fudusa diphologolo di le dikete mo tshireletsong e ntšha, go akaretsa malapa a tletseng a ditlou. Tshireletso e ne e le tsela ya go busetsa diphologolo kwa gae.

Mo thapama, mokoloko wa dintsa tsa naga o ne wa tshela tsela ka mokgwa wa go tshabela. Di ne di na le matshwao jaaka seatla se se tletseng pente, le ditsebe tse dikgolo tse di lekanyeng. Motsamaisi a bolelela bana gore di tsoma e le nngwe, le gore di abelana dijo le dintsintsi le baleti ba ba tlogetsweng mo bonnong jwa tsone. Ke nngwe ya diphologolo tse dikgolo tse di tshabelang go nyelela mo Aforika, a bua, mme Madikwe ke felo go sireletsegileng go tsone.

Dintsa di ne tsa tswa, tsa retologa ka botlalo mme tsa letela tse dingwe. Ya ntle, retologa, letela. Ya ntle, retologa, letela.

Kwa gae Neo a dira gore tlhware ya gagwe ya pampiri e dire jalo: tlola, retologa ka botlalo, letela. Nako nngwe le nngwe fa a e batla, o ne a tshwanetse go aga tse tharo gape. Lesedi a re, "E fe leina." Ba e bitsa "pack run". Jaanong leina le le lengwe le ne le tshwere dikgato tse tharo.

Potso ya kgang: *Tatelano efe e e boeletswang e e tshwanetseng go nna le leina la yone?*

Bokgoni: Dibolokwa tse di rulaganyeditsweng

Dira

1. Okeletsa sprite ya tlhware ya pampiri. Mo Dibolokong tse dingwe, dira boloko jo bo ikgethileng jo bo bidiwang 'pack run'. Fitlha tlhaloso ya jone mme o gokelele Tlola 20, Retolola 180 ka fa molemeng, morago Ema 1 ka fa tlase ga yone.
2. Mo setakeng se segolo sa tlhware, baya Boeletsa 2 ka fa tlase ga Play. Baya boloko jo bo ntšhwa jwa pack-run mo teng ga Boeletsa.
3. O se ka wa baya tlhaloso mo teng ga setaki se segolo. E tlhalosa se boloko jo bo ntšhwa bo se dirang fa bo bidiwa.

Khoutu (recipe)

```
TLHWARE YA PAMPIRI: GO TLHALOSA
Define pack run
Jump 20
Turn clockwise 180
Wait 1 second
```

```
TLHWARE YA PAMPIRI: GO DIRISA
When Play starts
Repeat 2 times:
    pack run
```

Go dira jang

Boloko jo bo ikgethileng bo naya tatelano leina. Go bo bitsa go dira ditiro tse di mo tlhalosong ya jone. Mo OctoStudio dibolokwa tse di ikgethileng ke tsa sprite e e di tlhamileng. Ga di bonale ka othomathiki mo go disprite tse dingwe tsotlhe.

Leka

Tobetsa Play: solofela go tlola gabedi le go retologa gabedi go go nang le go ema. Fetola go tlola mo tlhalosong mme o tobetse Play gape. Ditiriso tsa bobedi di tshwanetse go fetoga ka gonne di bitsa tlhaloso e le nngwe.

Fetola

Dira kgato ya bobedi e e rulagantsweng ka motsamao o o farologaneng. Dirisa maina a mabedi ka tatelano e e anelang tshimologo, bogare le bokhutlo.

Tlhalosa le go abelana

Letla molekane wa gago gore a tlhalose setaki se segolo a sa begele tiro nngwe le nngwe. Morago a bule tlhaloso go bona gore e dira jang.

PROJEKE 19

Kgang ya motho yo o kgakala



E ne e le beke ya bofelo ya ga Neo kwa Mahikeng. Mmaagwe o ne a tla mo tsaya ka Mosupologo. O ne a batla gore tsala ya gagwe Palesa kwa Johannesburg a bone tsotlhe. Fela Palesa o ne a se na OctoStudio, mme mogala wa gagwe o ne o tlhaela sebaka ka metlha.

Nkuku a botsa potso e e siameng. "A o tlhoka go lebelela, kgotsa o tlhoka go e fetola?" Neo a akanya. Fa Palesa a ne a batla go bona fela, rekoto e khutshwane e ne e tla lekana. Fa a ne a batla go tobetsa, go šikinya le go fetola mafoko a tlhware, o ne a tla tlhoka faele ya nnete ya porojeke le app.

Ba dira tsotlhe tse pedi. Vidio e nnye gore e romelwe bosigong jono. Le faele ya porojeke, e bolokilwe sentle, gore Palesa a e bule fa a ikemetse. Neo a kwala nokwane ka mokwalo wa gagwe mme a e tsenya mo kgetsaneng le bukana ya gagwe ya go taka. "Eno ke gore o e lebelele. Eno ke gore o e fetole."

Potso ya kgang: *Moreki wa gago o tlhoka go lebelela, kgotsa go bula le go fetola khoutu?*

Bokgoni: Vidio, GIF le faele ya porojeke

Dira

1. Bula kopi ya Porojeke 9 kgotsa porojeke e nngwe e khutshwane e e fedileng. Sekaseka setshwantsho sengwe le sengwe, lefoko le modumo le motho yo o godileng. Tlosa tshedimosetso ya poraefete pele ga go romela.
2. Bula Abelana > Rekota Vidio. Simolola go rekota mme o dire porojeke o dirisa dikgopolo tsa sekirini. Ema ka fa tlase ga

metsotswana e le 30. Dirisa kgetho ya go boloka ya sediriswa fa e le teng.

3. Leka Abelana > Dira GIF gore o nne le loop e khutshwane e e didimetseng, go fitlha metsotswana e le 15. Morago Abelana > Romela Faele ya Porojeke mme o boloke kopi ya .octostudio mo selegaeng. Ga o tlhoke go romela le fa e le efe ya tse kwa go mongwe.

Khoutu (recipe)

```
TIRAGATSO E E KHUTSHWANE YA GO REKOTA
Use your completed greeter or story project
Press Play and show one clear beginning and ending
Keep video under 30 seconds; GIF under 15
```

Go dira jang

Vidio e bontsha tiragatso e le nngwe e e rekotilweng mme e ka akaretsa modumo. GIF ke loop e khutshwane e e bonalang e e senang modumo. Faele ya porojeke e boloka porojeke e e ka baakanngwang, e e dirisanang mme e tlhoka OctoStudio gore e bulwe. Rekoto ga e boloke go tobetsa kgotsa go sega ga nako ya nnete.

Leka

Bula rekoto e e bolokilweng mme o lekole tiragatso yotlhe. Bula faele ya porojeke mo OctoStudio mme o lekole dibolokwa tsa yone. Fa letlapa la go abelana le farologane, kopa motho yo o godileng go batla tsela ya go boloka faele mo selegaeng go na le go e baya mo inthaneteng.

Fetola

Baakanya GIF e e didimetseng le vidio e e anelwang ya kgopolo e le nngwe. Bapisa gore ke bokao bofe jo bo sirelediwang ntle le modumo.

Tlhalosa le go abelana

Boloka kopi pele ga go tlosa app kgotsa go phimola data ya app. Go abelana mo inthaneteng go ka dirisa data mme go tlhoka tlhatlhobo ya babogedi le poraefete ya motho yo o godileng. Ga go na ngwana yo o tshwanetseng go gatisa tiro ya gagwe gore a fetse buka eno.

PROJEKE 20

Se re se isang kwa gae



Mo mantsiboeng a bofelo ba boela kwa Leseding la Molemane. Metsi a ne a le phepa jaaka letsatsi la ntlha. Nkuku a nna mo lefikeng le le sephara mme bana ba nna go mo dikologa, jaaka ba dira jaanong ka metlha.

Neo a bula bukana ya gagwe ya go taka. E ne e tletse. Tlhware kwa Leseding. Tsela go ya kwa metsing. Lerako la dipalesa tse di retologang. Setlhare se se golang. Lefika le le fitlhileng tlhware. Kgang e e rekotilweng. Pontsho kwa dam. Thaba kwa Taung. Kgaswane e e tsamayang. Dithaba mo dikganyeng tse tharo. Thobane e e fetisiwang go tswa mo seatleng. Ditlhase tsa mollo mo thabeng. Dikgakana tse tharo le kgaka. Mokoloko o tswa, o retologa, o boa.

Kwa lotlhakoreng la sediba go ne go le teng lekgapetla la polasetiki. Kagiso a le tsaya a le tsenya mo kgetsaneng. Neo a mmona mme a itse gore porojeke ya gagwe e ntšhwa e tla re eng.

E ne e dirisa tsotlhe. Moetapele wa tlhware yo o go amogelang, lekgapetla le o ka le tobetsang go thusa, le bokhutlo jo bo farologaneng fa o dira jalo. Fa e fetsa, Nkuku a bua mafoko a bogologolo: "Motho ke motho ka batho." Neo a tobetsa tlhware. "Dumela," ya re. "Dumela," ba bua botlhe, mme metsi a tswelela.

Potso ya kgang: *Phetogo efe e o ka ratang gore motho a e akanye mo setšhabeng sa gago?*

Bokgoni: Itlhame kgang ya gago e e dirisanang

Dira

1. Rulaganya kgang ya tlhokomelo ya lefelo ka metsotso e meraro mo pampiring: tikologo, sengwe se moreki a ka se dirang, le bokhutlo. Tlhopha ditshwantsho tsa ntlha kgotsa disprite tsa lebokoso la dibuka, metheo e mebedi e e amogetsweng le modumo o le mongwe o mokhutshwane wa kgetho.

2. Aga sedirisiwa sa tshimologo se se ka fa tlase ka sprite ya Moetapele wa Tlhware ya Pampiri le sprite ya Lekgapetla, lekgapetla la polasetiki kwa Leseding. Tlhopha molaetsa o le mongwe wa teng wa 'ready' le o mongwe wa 'ending'; di tshwanetse go farologana. Boloka Moetapele a bonala mo metheong yotlhe e mebedi.

3. Dirisa dikgetho tse di tshwanang tsa go romela/go amogela. Okeletsa mafoko a gago le karolo e le nngwe go tswa mo porojekeng ya pele: diaparo, boloko jo bo ikgethileng, mmala kgotsa go boeletsa go go balwang. Boloka karolo eo e le khutshwane go lekana gore e leke.

Khoutu (recipe)

```
MOETAPELE: GO AMOGELA
When Play starts
Switch to welcome backdrop (instant)
Say: Tap the scrap to help. for 3 seconds
Send READY message to all
```

```
LEKGAPETLA: GO BUSETSA LE GO BONTSHA
Stack 1: When Play starts > Hide (instant)
Stack 2: When READY message received > Show (instant)
```

```
LEKGAPETLA: GO TOBETSA GO FETSA
When this sprite tapped
Hide (instant)
Send ENDING message to all
```

```
MOETAPELE: BOKHUTLO
When ENDING message received
Switch to second backdrop (instant)
Say: Together we can care for our home. for 3 seconds
```

Go dira jang

Lekgapetla le na le ditaki tse tharo tse di kgaoganeng: Play > Fitlha; Amogela READY > Bontsha; Tobetsa Lekgapetla > Fitlha > Romela ENDING. Setaki sa Play se se busetsa. Melaetsa e rulaganya kgang mo disprite tsotlhe. Dirisa kgetho e e tshwanang ya molaetsa kwa go romela le go amogela ngwe le ngwe e e tshwanang.

Leka

Tshameka go tswa kwa tshimologong: lekgapetla le fitlhilwe fa go amogelwa, le a bonala morago ga READY, morago go tobetsa go le lengwe go fetola lefelo mme go le fitlha. Simolola gape mme o boeletse. Lekola dikgetho tsa melaetsa le dibolokwa tsa go busetsa pele ga go okeletsa dikarolo tse dingwe.

Fetola

Dira sepheo e le sa gago, e seng fela motshameko wa matlakala. Letla tsala gore e leke ntle le go e kaela. Lebelela gore a ema kae, botsa gore ke eng se se mo tlhakantseng, mme o fetole selo se le sengwe.

Tlhalosa le go abelana

Boloka kopi e e ka baakanngwang. Bontsha kgang, tlhalosa ketsakgato e le nngwe le loop e le nngwe kgotsa tlhopho, mme o bolele phetogo e o e dirileng morago ga go leka. Tlhopha go romela go go sireletsegileng fela morago ga tlhatlhobo ya motho yo o godileng.

Dira gore e dire mo go mongwe

Leka, lebelela, baakanya

Tshameka go tswa kwa tshimologong: lekgapetla le fitlhilwe fa go amogelwa, le a bonala morago ga READY, morago go tobetsa go le lengwe go fetola lefelo mme go le fitlha. Simolola gape mme o boeletse. Lekola dikgetho tsa melaetsa le dibolokwa tsa go busetsa pele ga go okeletsa dikarolo tse dingwe.

Dira e nne ya gago

Dira sepheo e le sa gago, e seng fela motshameko wa matlakala. Letla tsala gore e leke ntle le go e kaela. Lebelela gore a ema kae, botsa gore ke eng se se mo tlhakantseng, mme o fetole selo se le sengwe.

Bolelela mongwe

Boloka kopi e e ka baakanngwang. Bontsha kgang, tlhalosa ketsakgato e le nngwe le loop e le nngwe kgotsa tlhopho, mme o bolele phetogo e o e dirileng morago ga go leka. Tlhopha go romela go go sireletsegileng fela morago ga tlhatlhobo ya motho yo o godileng.

Go akanya ga moitirelo wa gago

Kgang ya me e ka ga ___. Moreki wa me a ka ___. Ketsakgato e e simololang porojeke ya me ke ___. Khoutu ya me e gopola kgotsa e boeletsa ___. Ka tshimologo ke ne ke solofetse ___. Morago ga go leka, ke fetotse ___. Kgopolo ya me e e latelang ke ___.

Tlhatlhobo e e utlwelang botlhoko ya molekane

Botsa: Ke motsotso ofe o o dutseng le wena? A go ne go na le karolo e o neng o sa kgone go e bona kgotsa go e dirisa? A dikheshene, sprite e kgolo kgotsa tsela e nngwe ya go tsenya di ka thusa? Neela kakgelo e le nngwe e e rileng, e seng dintlha tsa gore ke mang yo o neng a potlaka.

Fa disediriswa di farologana

Ga go na ngwana yo o tlogelwang kwa ntle ga porojeke

Khamera e ganetswa? Penta selo. Maekerofouno e ganetswa? Dirisa modumo wa lebokoso la dibuka le kheshene. Ga go na go sega? Dirisa tsela e nngwe ya disprite tsa metshwari mo Porojekeng 14. Ga go na go šikinya? Dirisa setaki sa go tobetsa mo Porojekeng 13. Mo Chromebooks dirisa taolo ya go sega ya sekirini le dikonope tsa go šikinya kgotsa magnete; o se ka wa šikinya laptop ka mmele.

Magnete, go tshikinyega le lebone ke dikgetho

Kaedi ya semmuso e na le boloko jwa go simolola ka magnete, fela disediriswa dingwe tsa Android ga di bo tshegetse. E na le le go tshikinyega: ke mogala o o nang le haptic feedback fela o tshikinyegang, disediriswa tse dingwe di letsa modumo wa buzz. Boloko jwa flashlight bo tlhoka hardware e e siameng mme bo ka gotlhola sethale go na le moo. Ga go na se se tlhokegang mo bukeng eno. O se ka wa dirisa dimagnete tse di nonneng, dikganya tse di panyang kgotsa didiriswa tse di nonofileng mo seetong sa bana.

Beam ga se molaetsa wa poraefete

Beam e romela letshwao la Bluetooth kwa disediriswa tsa OctoStudio tse di gaufi; e farologane le go Romela/Amogela mo porojekeng e e dirisitsweng mo Diporojekeng 11 le 20. Kaedi e re di-phone le di-tablet di ka romela; Chromebooks di ka amogela mme ga di kgone go romela. Disetingi tsa Bluetooth le ditetla di a e ama. Tsela e thusa go kgaoganya ditlhopha mme ga se kano ya sephiri.

Pontsho ya motho yo o godileng fela, ke kgetho: mo mogaleng o le mongwe kgotsa tablet, simolola ka Tobetsa > Beam letshwao le le tlhophilweng mo tseleng e e tlhophilweng. Mo sedirisweng se se gaufi dirisa Fa beam eo e amogelwa > bubble ya puo. Tshwantsha letshwao le tsela, bula Bluetooth ka tetla, mme o leke mo sekgaleng se se gaufi. O se ka wa akaretsa data ya poraefete. Fa e retelelwa, boela kwa tirong ya molaetsa wa mo porojekeng.

Tiro ya hardware le go tsena go tshwanetse go lekolwa mo disedirisweng tse di dirisiwang tota. Ga go na tiro e e ikanyang ka Bluetooth, magnete, lebone kgotsa go tshikinyega. Go dira ntle le inthanete pele ga go reye gore sediriswa sengwe le sengwe se na le sensara nngwe le nngwe.

Go dira ga ntlha ga motho yo o thusang

Pele ga go ruta tsotlhe tse 20

- Lekola dikarolo tsa nnete tsa disediriswa, dikgatiso tsa app, puo, sebaka le ditetla. Dirisa sediriswa se le sengwe sa Android kgotsa iOS ka bobedi fa go kgonega; akaretsa diphetogo tsa Chromebook fa e le dilo tsa gago.

Ruta ka dikgato tse khutshwane tsa go dira

Bala kgang ka lentswe le legolo ntle le go e kgaoganya go bua maina a dibolokwa. Botsa potso e le nngwe e e nang le bokao. Bontsha tiro e le nngwe e ntšhwa ya sekirini, morago o letle babedi gore ba e leke. Neela nako ya go fetola setshwantsho le mafoko. Fetsa ka go bontsha, go tlhalosa le go boloka kopi, e seng lobelo la go fetsa. Seno ke diporojeke, e seng lenaneo le le tlhomameng la porojeke e le nngwe ka beke.

Pele ga go ntsha: lekola kutlwisiso ya bana, ditshwanelo tsa media, tiriso e e sireletsegileng le go balega ga kgatiso. O se ka wa bolela gore buka eno e amogetswe ke DBE kgotsa gore e na le mmapa o o feletseng wa CAPS.

Bona se moithuti a se tlhaloganyang

Reetsa ditlhaloso, e seng fela dipholo

Dirisa ditlhaloso tse nne: "Ke e lekile ka thuso", "Ke a e kgona", "Ke ka e tlhalosa", "Ke ka e fetolela mongwe". Ke ditlhotlheletso tsa tshegetso, e seng maina a a kgomaretswang ngwana. Kwala fela se se tlhokegang mo phaposing; boloka direkoto tsa motho ka bongwe e le sephiri.

Morago ga 1-5

Bona Gae le moeditara. Tlhopha sprite e e siameng. Tshwantsha ketsakgato ya go simolola le tiro. Akanya go boeletsa. Busetsa boleng jwa tshimologo.

Morago ga 6-10

Tlhopha media ka tetla. Tlhalosa phapang gareng ga sprite le motheo. Dira diaparo tse pedi tse di tshwanang. Tlhalosa phetogelo ya lefelo.

Morago ga 11-14

Tshwantsha molaetsa wa mo teng. Fa tsela e nngwe ya go tsenya ka sensara. Dirisa loop go tsamaisa ka metlha. Leka ka disprite tse pedi.

Morago ga 15-18

Busetsa le go fetola bolokwa se se fetogang. Tlhalosa tlhopho e le nngwe. Lemoga gore tlhopho e e sa rulagangwang e ka boeletsa. Dirisa le go tlhaloganya boloko jo bo ikgethileng.

Morago ga 19-20

Tlhopha gareng ga faele e e ka baakanngwang le rekoto. Tlosa dilo tsa poraefete. Tlhalosa porojeke. Tlhalosa tokafatso e le nngwe e e dirilweng morago ga fa motho yo mongwe a e lekile.

STEAM e mo tirong

Saense: lebelela dintlha tsa setshwantsho mme o bapise ditebelelo le dipholo. Theknoloji: tlhaloganya phapang ya media le disediriswa. Boenjiniere: rulaganya, aga, leka le go fetola tirisano. Bonang: taka, ruta, anela le go tlhama mo babogedi. Dipalo: bala ditotololo, diphetogo, dibophara le dipheo. Tlhokomelo ya batho, mafelo le bosupi jo bo tshelang e naya diporojeke maikaelelo a a abelanwang; bonang bo tsholetsa bokao jwa tsone.

Tlhaloso ya gago e nnye ya mafoko

Motheo / Lefelo le le kwa morago ga disprite.

Boloko / Taelo e le nngwe kgotsa boleng bo le bongwe mo app.

Tlhopho / Teko e e nang le karabo ya nnete kgotsa maaka.

Seaparo / Sebopego se sengwe kgotsa maemo a sprite e le nngwe.

Boloko jo bo ikgethileng / Tatelano e e filweng leina e o e tlhalosang mo sprite e le nngwe.

Ketsakgato / Sengwe se se simololang setaki: play, go tobetsa, molaetsa kgotsa sengwe se se tshwanang.

Romela kwa ntle / Dira kopi kwa ntle ga app, jaaka vidio kgotsa faele ya porojeke.

Loop / Khoutu e e boeletsang. Boeletsa e a bala; Ka metlha e tswelela go fitlha e emisiwa.

Molaetsa / Letshwao gareng ga disprite. Beam ke letshwao le le farologaneng la disediriswa tse di gaufi.

Morepoti / Boloko jo bo ntshitsweng jo bo naya boleng mo teng ga boloko jo bongwe.

Busetsa / Naya porojeke boleng jwa yone jwa tshimologo gape.

Tatelano / Ditaelo tse di rulagantsweng ka tatelano.

Sprite / Sebopego, setshwantsho kgotsa selo se o ka se kwalelang khoutu.

Setaki / skripte / Dibolokwa tse di golaganeng tse di dirang mmogo.

Sethale / Lefelo kwa kgang e bonalang teng.

Bolokwa se se fetogang / Leina le le bolokang palo e porojeke e ka e fetolang.

Mafoko a mararo a go a boloka

Rulaganya. Leka. Fetola. O ka se tlhoke gore kgopolo ya gago ya ntlha e nne ya bofelo.

Mothopo le ditshwanelo (ka Seesemane): letlhare le le latelang le le kwa tlase le kwadilwe ka Seesemane gore ditsopolo di nne di sa fetolwe. Dikgang di tlhamilwe tsa ntlha; ga se phetolelo.

Sources and credits

OctoStudio was created by the Lifelong Kindergarten group at MIT Media Lab. To learn more, visit octostudio.org. This is an independent Circuit Kids companion book, not an OctoStudio or MIT publication or endorsement.

Stories, teaching text and schematic diagrams in this book were written for Circuit Kids. The Coding Cards section at the back reproduces the official OctoStudio Coding Cards v3.0 (August 2026) unchanged, under CC BY-NC-SA 4.0, with the credit: OctoStudio was created by the Lifelong Kindergarten group at MIT Media Lab. The code recipes describe documented functions in original project contexts. Original book material is licensed under CC BY-NC-SA 4.0 (see page 2). Production-rights clearance remains a release gate.

Current Reference Guide, July 2026, v1.1.0+ https://resources.octostudio.org/ReferenceGuide.pdf

Reference-guide landing page https://octostudio.org/support/learning-resources/octostudio-reference-guide

Learning resources https://octostudio.org/learn/

OctoStudio in Schools https://octostudio.org/learn/octostudio-in-schools

Bring a Book to Life https://octostudio.org/learn/bring-a-book-to-life

Age guidance: especially ages 7-12 https://octostudio.org/support/faq/what-age-group-is-octostudio-designed-for

Offline guidance https://octostudio.org/support/faq/do-i-need-an-internet-connection-and-data-to-use-octostudio

App privacy policy https://octostudio.org/privacy-app

Primary sources: official documentation, official educational resources and published policy/FAQ. No physical-device trial was possible for this edition. Older search results can surface the July 2025 guide, which omits newer costume features.

Where the stories come from

Places and facts in the stories come from public sources: Taung Child, https://humanorigins.si.edu/evidence/human-fossils/fossils/taung-child ; the Pilanesberg ring dyke complex, https://www.pilanesbergnationalpark.org/media/volcano-gave-birth-game-reserve/ ; Madikwe Game Reserve and Operation Phoenix, https://experiencemadikwe.com/content/page/wildlife-and-conservation.html ; Kgaswane Mountain Reserve,

https://parksnorthwest.co.za/kgaswane/Kgaswane%20brochure.pdf ; the Molemane Eye, https://sahikes.com/southern-africa-hikes/north-west/molemane-eye-nature-reserve . The Kwanyape whirlwind in Story 17 is told as an old story, from Bakgatla oral tradition recorded by Schapera, https://www.lacerta.de/AF/Bibliografie/BIB_10772.pdf . Characters, journeys and conversations are fiction. No venue or community endorsement is claimed.

Rights and testing notes

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The OctoStudio Coding Cards v3.0 PDF (August 2026) states CC BY-NC-SA 4.0 and allows translation, and this book includes them in that spirit with the required credit. Its landing page has been seen to state Attribution-ShareAlike, so ask the publisher to clarify before using the cards in a paid print bundle.

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Coding Cards source: https://octostudio.org/support/learning-resources/octostudio-coding-cards and https://resources.octostudio.org/OctoStudio-Coding-Cards.pdf

Learning and print checks

- This edition is for ages 7-12, with a North West family and a paper python that travels with them.
- Project 20 retains an extra testing and reflection section.
- Read and make together. Talk about the stories, care for places, and the difference between new fiction and traditional accounts.
- Test projects on the devices you will use before teaching a group. Check the expected results and adapt the pace to your learners.
- CC BY-NC-SA 4.0 applies to this book. Keep the source-rights record with it.

Circuit Kids | Registration in progress | 072 123 1490 | info@circuitkids.co.za

Dikarata tsa Khoutu

Dikarata tsa Khoutu tsa OctoStudio tse di semmeng di bontsha gore bloko le setlhopha sengwe le sengwe di dira jang. Di dirisediwe gaufi le dipheti tsa diporojeke.

Dikarata di fa fa tlase ga laesense ya Creative Commons Attribution-NonCommercial-ShareAlike 4.0 (CC BY-NC-SA 4.0), e e letlelelang go ranolwa. Ditshwantsho tsa dikarata ke tsa dikarata tsa ntlha; dipolelo tse di kwadilweng fa tlase ga tsone ke tsa phetolelo. Maina a dibloko a nna ka Seesemane jaaka a bonala mo OctoStudio.

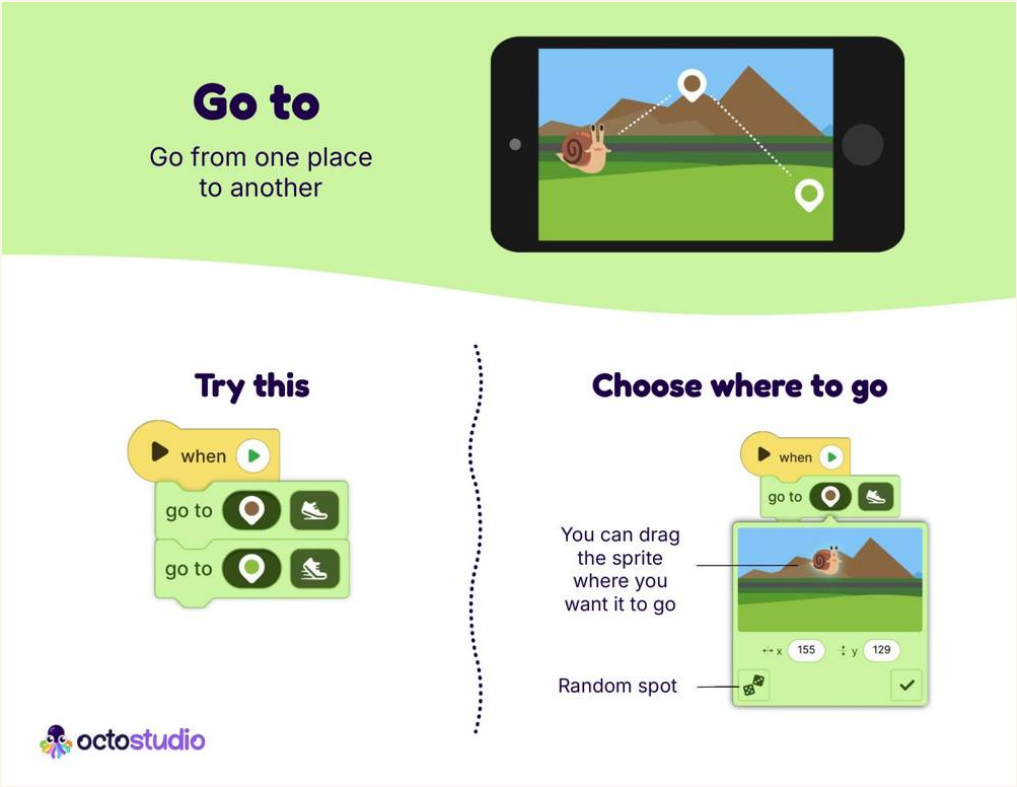
OctoStudio was created by the Lifelong Kindergarten group at MIT Media Lab.

Cards v3.0, CC BY-NC-SA 4.0.

Cards v3.0 (August 2026) - https://octostudio.org/support/learning-resources/octostudio-coding-cards

Ya kwa

Tswa kwa lefelong le le lengwe o ye kwa le lengwe

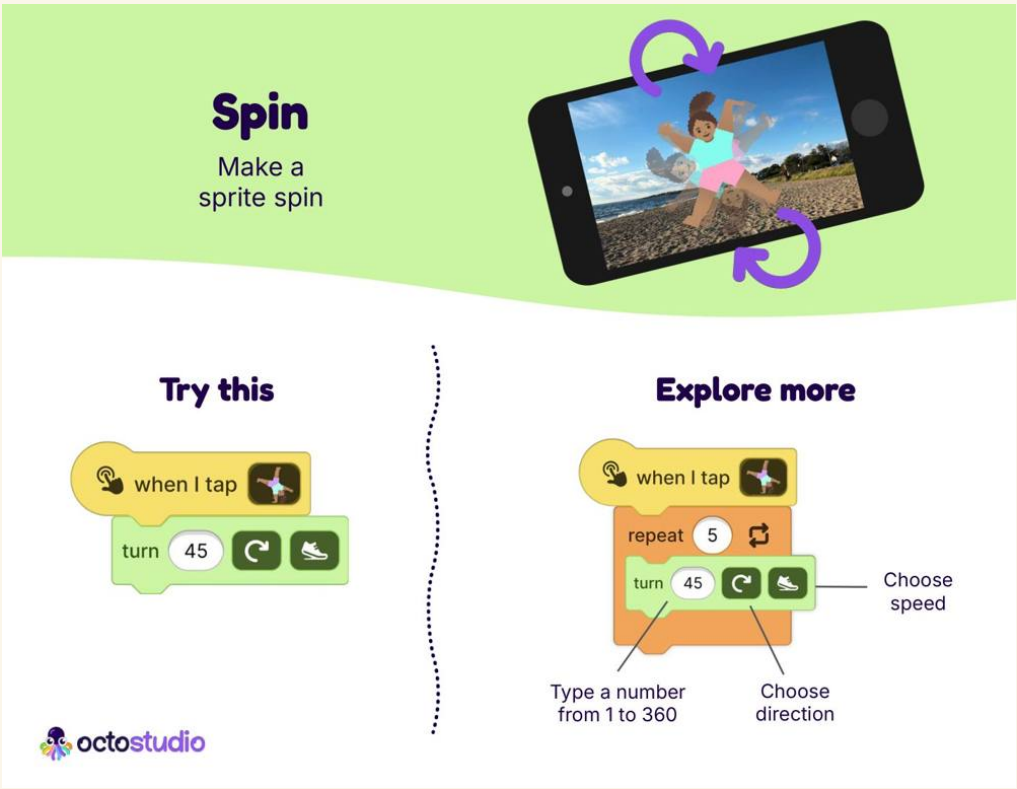


Karata ya ntlha ya OctoStudio (Seesemane).

- Leka seno
- Tlhopha gore o ya kae
- O ka goga sprite go ya kwa o batlang gore e ye teng
- Felo fa go sa rulaganngwang

Retologa

Dira gore sprite e retologe

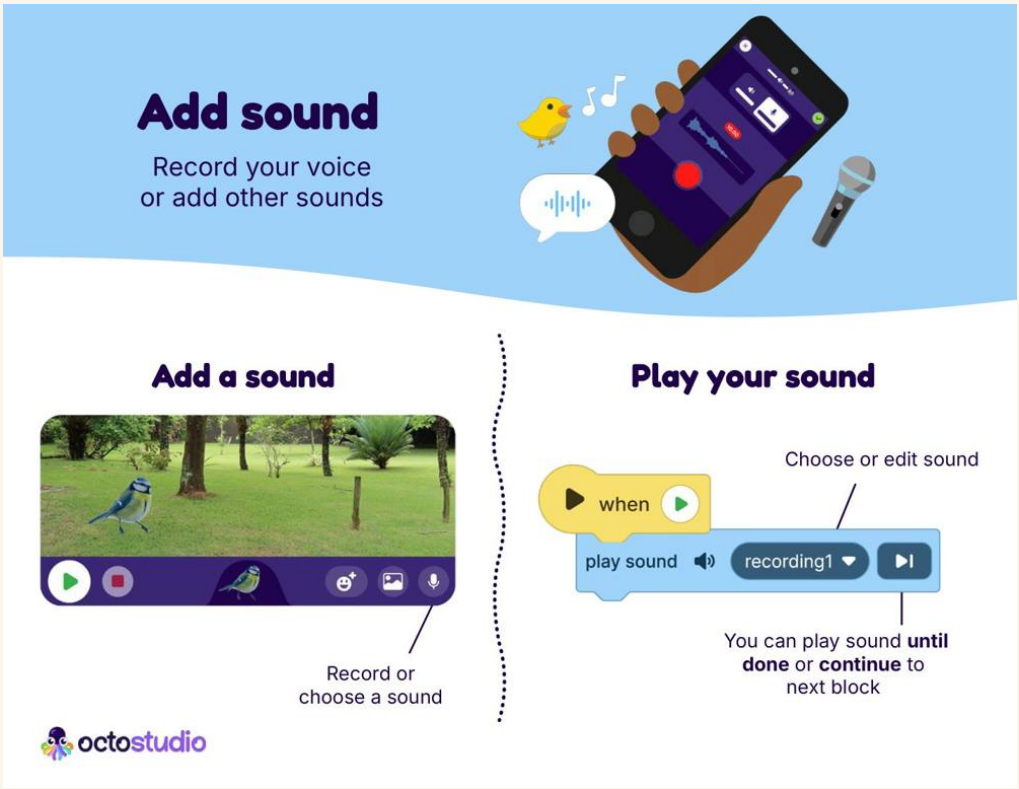


Karata ya ntlha ya OctoStudio (Seesemane).

- Leka seno
- Batlisisa go feta
- Tlhopha lebelo
- Kwala palo go tswa kwa 1 go ya kwa 360
- Tlhopha ntlha

Okeletsa modumo

Rekota lentswe la gago kgotsa o okeletse medumo e mengwe

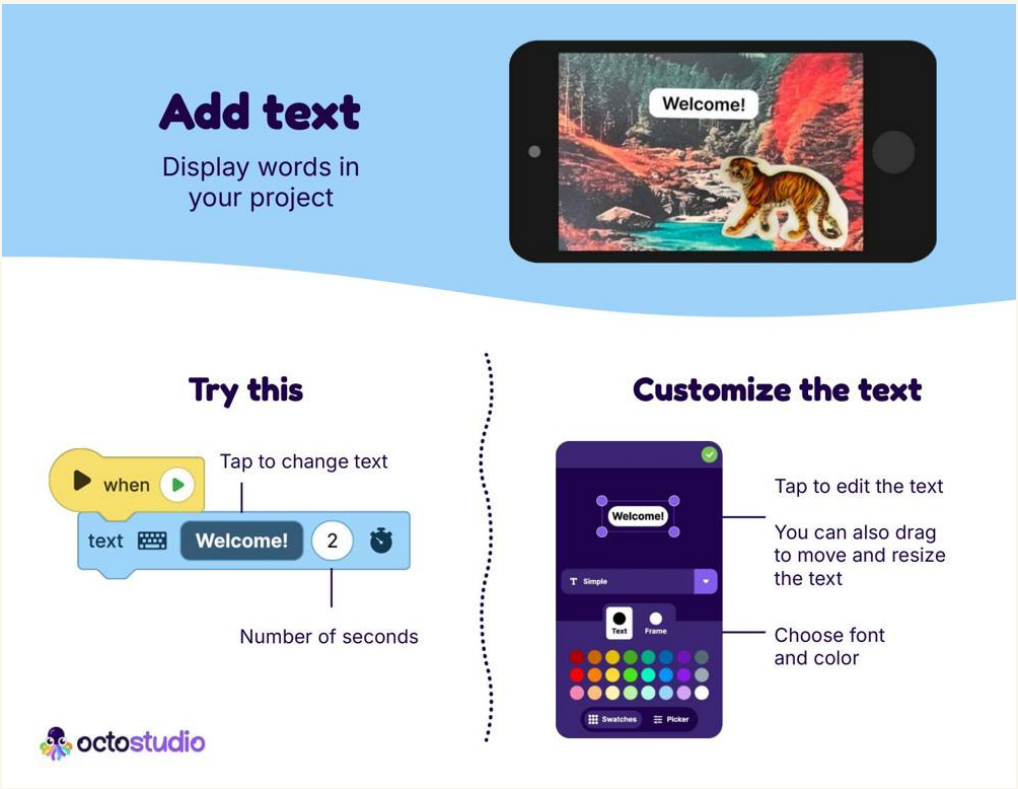


Karata ya ntlha ya OctoStudio (Seesemane).

- Okeletsa modumo
- Letsa modumo wa gago
- Tlhopha kgotsa o baakanye modumo
- Rekota kgotsa o tlhophe modumo
- O ka letsa modumo go fitlha o fela kgotsa o tswelele kwa boloko jo bo latelang

Okeletsa mokwalo

Bontsha mafoko mo porojekeng ya gago

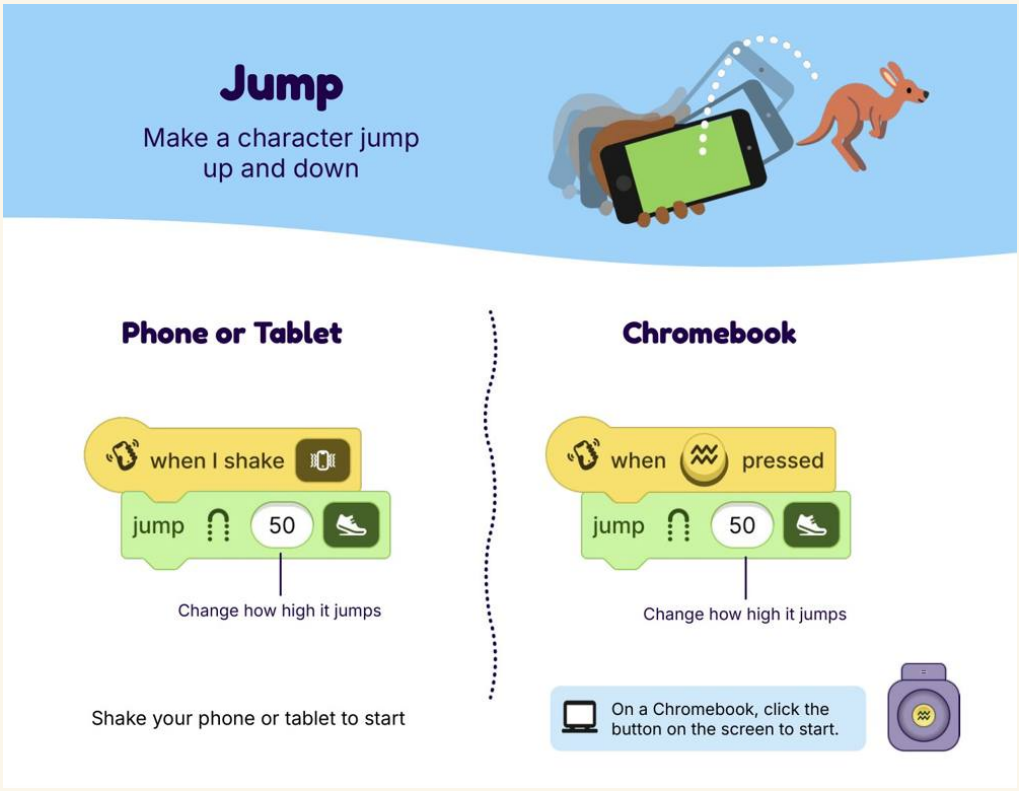


Karata ya ntlha ya OctoStudio (Seesemane).

- Leka seno
- Baakanya mokwalo
- Tobetsa go fetola mokwalo
- Tobetsa go baakanya mokwalo
- O ka goga gape go o tsamaisa le go fetola bogolo jwa one
- Palo ya metsotswana
- Tlhopha mofuta wa letlhaka le mmala

Tlola

Dira gore sebopego se tlole kwa godimo le kwa tlase



Karata ya ntlha ya OctoStudio (Seesemane).

- Mogala kgotsa Tablet
- Fetola gore e tlola kwa godimo go le kae
- Šikinya mogala kgotsa tablet ya gago go simolola
- Chromebook
- Mo Chromebook, tobetsa konope e e mo sekirining go simolola.

Gola le go fokotsega

Dira gore sprite e gole kgotsa e fokotsege

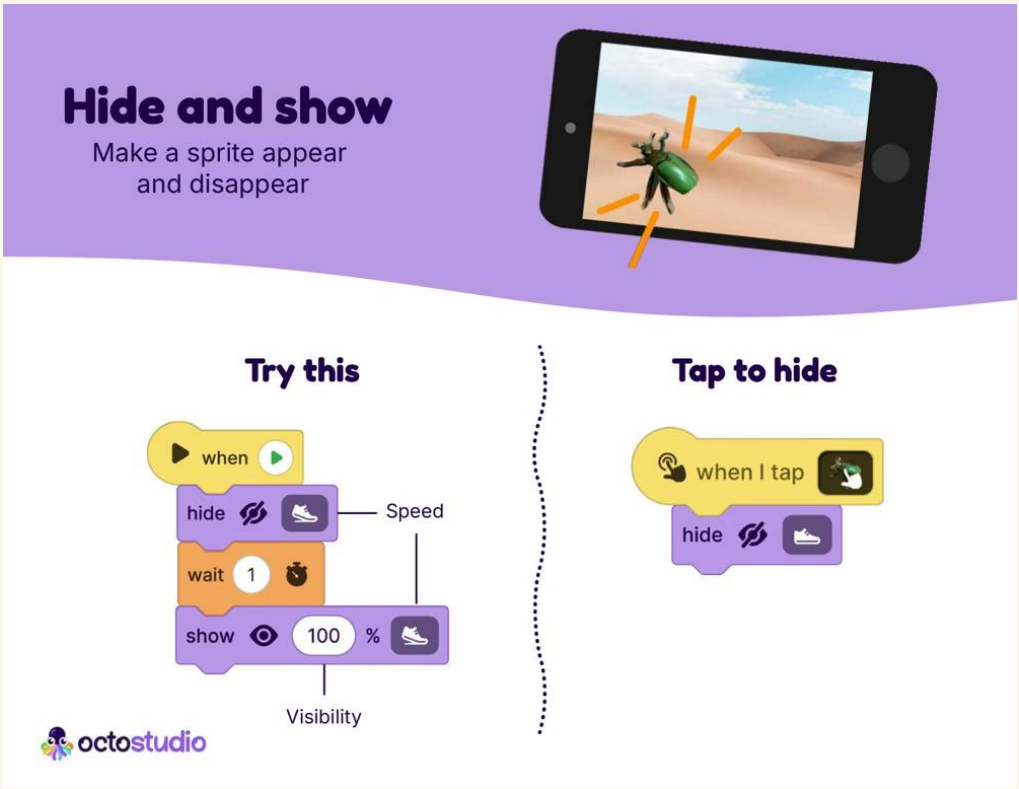


Karata ya ntlha ya OctoStudio (Seesemane).

- Leka seno
- Leka dipalo tse di farologaneng
- Okeletsa go feta

Fitlha le go bontsha

Dira gore sprite e bonale e bo e nyelele

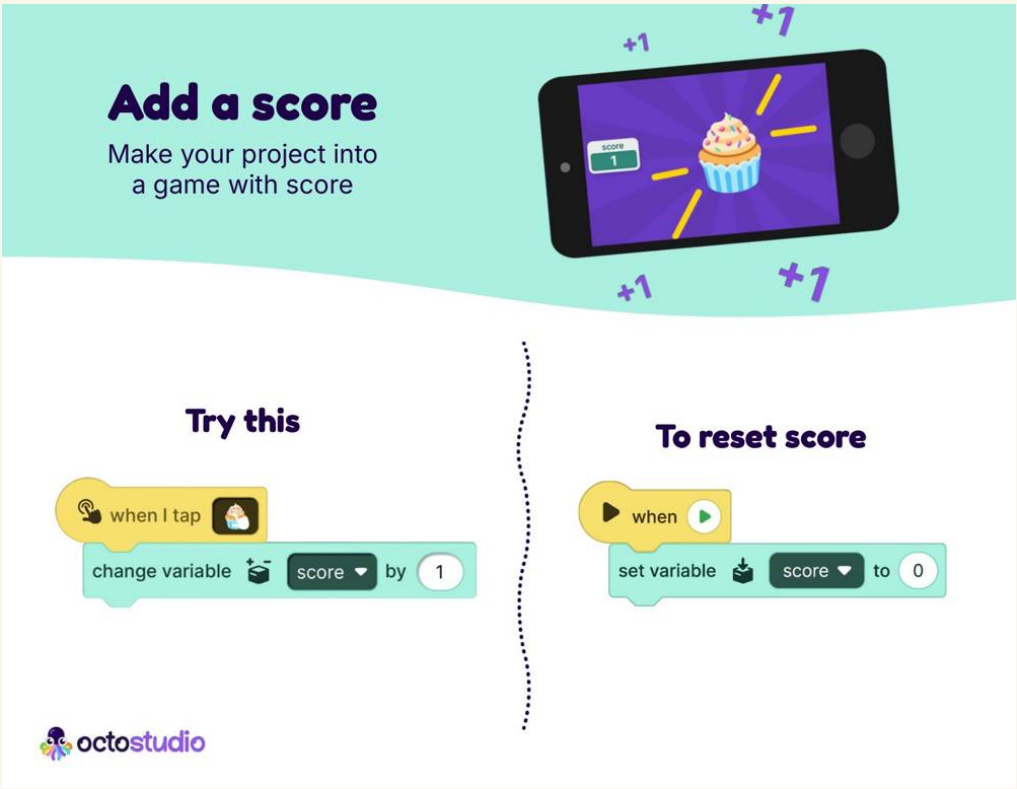


Karata ya ntlha ya OctoStudio (Seesemane).

- Leka seno
- Tobetsa go fitlha
- Lebelo
- Go bonala

Okeletsa dintlha

Dira porojeke ya gago motshameko o o nang le dintlha

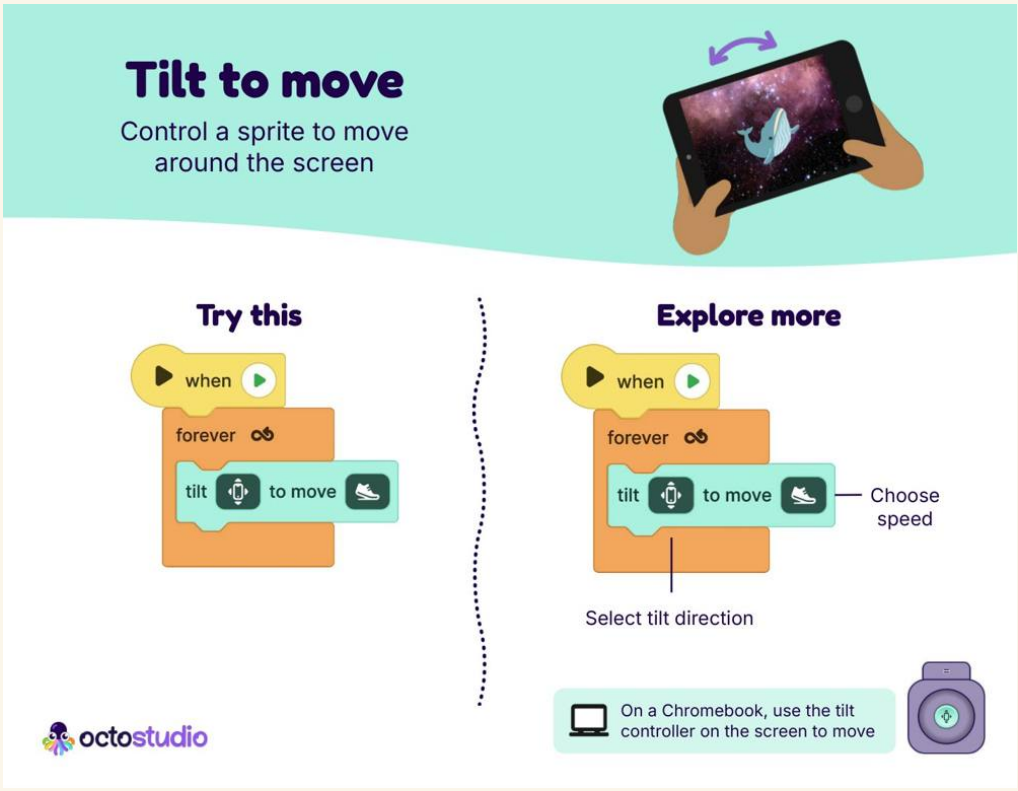


Karata ya ntlha ya OctoStudio (Seesemane).

- Leka seno
- Go busetsa dintlha

Sega go tsamaya

Laola sprite gore e tsamaye mo sekirining

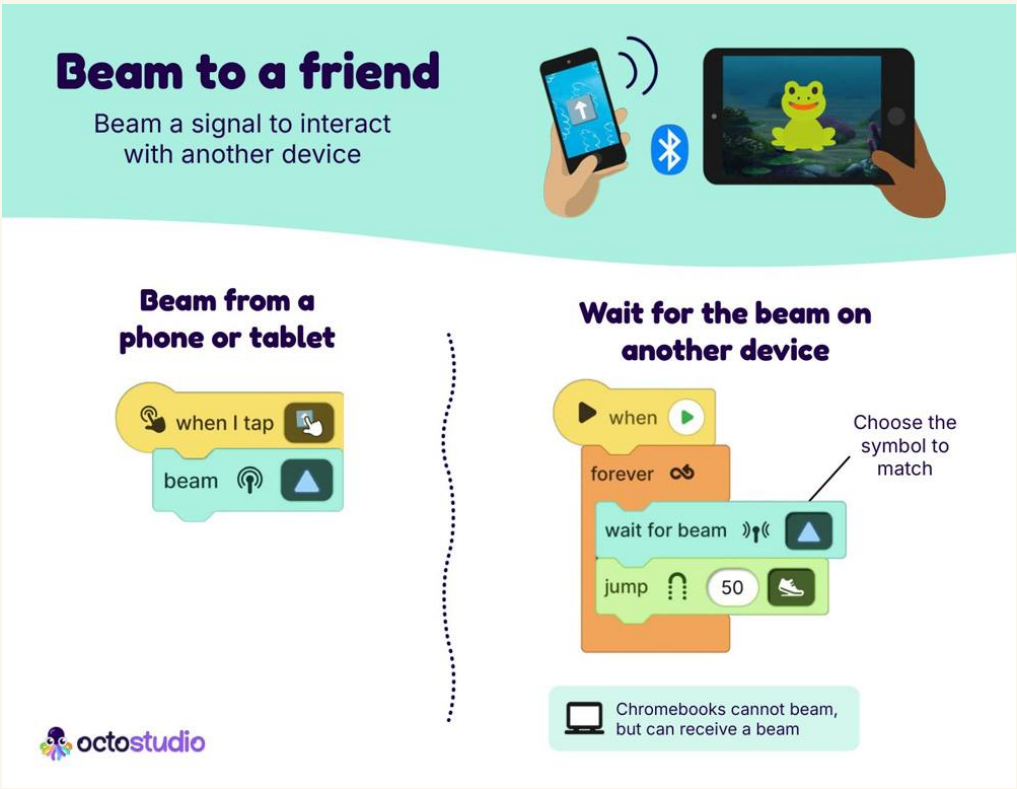


Karata ya ntlha ya OctoStudio (Seesemane).

- Leka seno
- Batlisisa go feta
- Tlhopha lebelo
- Tlhopha ntlha ya go sega
- Mo Chromebook, dirisa taolo ya go sega ya sekirini go tsamaya

Beam kwa tsaleng

Romela letshwao go dirisana le sediriswa se sengwe



Karata ya ntlha ya OctoStudio (Seesemane).

- Beam go tswa mo mogaleng kgotsa tablet
- Ema beam mo sedirisweng se sengwe
- Tlhopha letshwao le le tshwanang
- Chromebooks ga di kgone go beam, fela di ka amogela beam

Baya seaparo

Fetola go tswa mo seapareng se sengwe go ya kwa go se sengwe

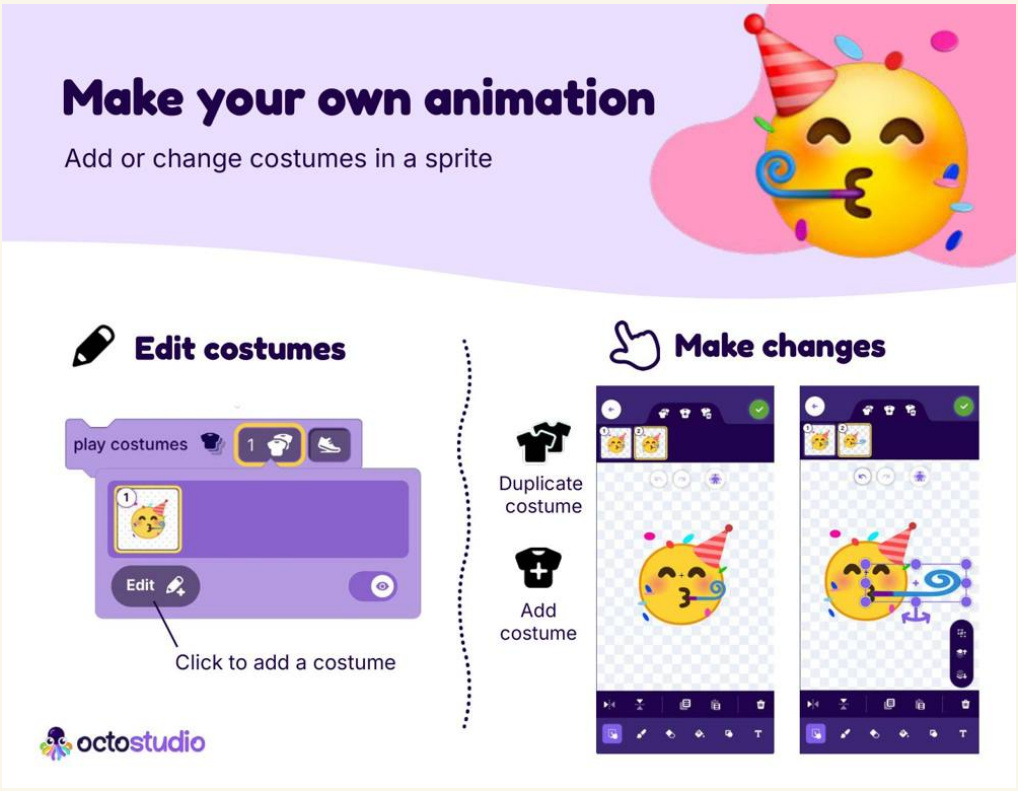


Karata ya ntlha ya OctoStudio (Seesemane).

- Bula lenaneo go tlhopha seaparo
- Fetola go tswa mo seapareng se sengwe go ya kwa go se sengwe
- Tlhopha lebelo

Dira motsamao wa gago o o bonalang

Okeletsa kgotsa o fetole diaparo mo sprite

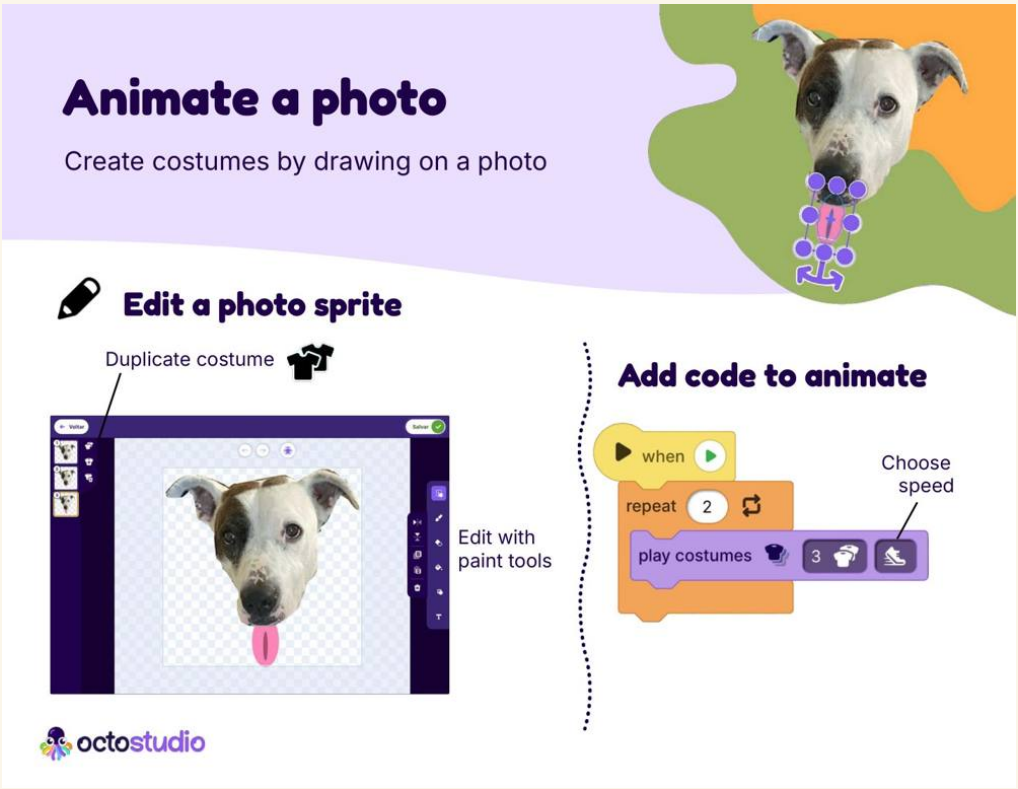


Karata ya ntlha ya OctoStudio (Seesemane).

- Dira diphetogo
- Baakanya diaparo
- Kopa seaparo
- Okeletsa seaparo
- Tobetsa go okeletsa seaparo

Dira setshwantsho se tsamaye

Dira diaparo ka go taka mo setshwantshong



Karata ya ntlha ya OctoStudio (Seesemane).

- Baakanya sprite ya setshwantsho
- Kopa seaparo
- Okeletsa khoutu go dira motsamao
- Tlhopha lebelo
- Baakanya ka didiriswa tsa pente

Motsamao

Dira gore sprite e tsamaye, e fofe kgotsa e rolege



Karata ya ntlha ya OctoStudio (Seesemane).

- Tlhopha sprite e e nang le diaparo
- Dira motsamao fa e ntse e tsamaya
- Diphologolo le batho ba le bantsi ba sprite ba tla le seaparo se le sengwe go feta.
- O ka dirisa ditaki tse pedi kgotsa go feta tsa dibolokwa go simolola ka nako e le nngwe.
- Tlhopha lebelo